

Particulate Matter

Week in Review

Week ending Saturday 1 August 2026 | pooled 27 July 2026 to 1 August 2026 | 131 records from 6 daily issues

131

RECORDS POOLED
THIS WEEK

20

SENSING RECORDS
(15% OF WEEK)

10

SUBTOPICS
ACTIVE

48

POOLED EFFECT
ESTIMATES

4

TRIALS ON
THE WATCH LIST

The week in one paragraph

Four independent papers this week attacked the exposure *variable* rather than the health model — and all four found it wanting. The through-line is that $\text{PM}_{2.5}$ mass at a fixed monitor is now the weakest link in the causal chain, and the field has started saying so with numbers.

Nyoni et al. (31 Jul, *Air Qual Atmos Health*) validated road-geometry proxies against a year of filter-based personal $\text{PM}_{2.5}$ in 343 postpartum participants across three sub-Saharan countries: weighted road density correlated only in Mozambique ($\rho = 0.351$) and **distance-to-highway inverted sign in Kenya** ($\rho = -0.224$, $p = 0.018$). Saeedi et al. (1 Aug, *ES&T*) showed land-use regression for mobile ultrafine particles falling from **~217% average percentage error under random cross-validation to ~79% under spatiotemporal CV** — the model was never predicting, the validation was. Blanco et al. (1 Aug, *JESEE*) resampled on-road campaign designs against a 5,283-person cohort and found every alternative design attenuated the health estimate. And Zhu et al. (31 Jul) reported pneumonia-admission RRs of **1.004 (0.991–1.017) for $\text{PM}_{2.5}$ against 1.221 (1.174–1.271) for NO_2** — PM mass was the null term in its own model.

What would have to be true for this to be wrong: the four results would have to be selection rather than signal. Two of them (Saeedi, Blanco) are methods papers whose premise is that current practice is inadequate, so they are not independent of the conclusion; Nyoni's setting is one where household biomass dominates the personal exposure budget, which is exactly where traffic geometry should fail; and Zhu's is a single-pollutant model over correlated combustion tracers, so it is a statement about correlation structure, not about which pollutant acts. **This is a real pattern, but it is a pattern about measurement adequacy, not about PM being harmless.** The distinction matters and the week's own papers do not always draw it.

Subtopic trend and inflection

Subtopic movement across the week (2026-07-27 to 2026-08-01)

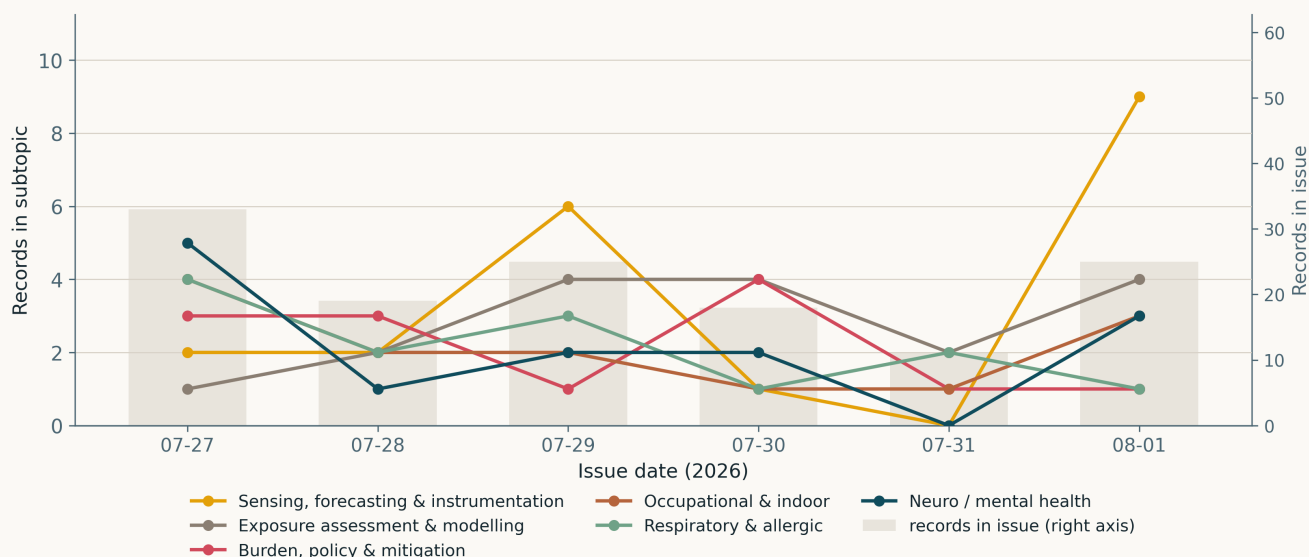


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **The one real inflection is sensing/instrumentation: 2, 2, 6, 1, 0, 9 across 27 July–1 August, against daily totals of 33, 19, 25, 18, 11, 25.** Both ends of that series are pipeline artefacts, not literature. The 6 on 29 July came from a Consensus backfill; the 0 on 31 July and the 9 on 1 August are the same defect and its fix — the Crossref by-journal leg was activated on 1 August and back-swept the week, so nine records that were deposited between 26 July and 1 August all landed in one issue. **Read the weekly total (20/131, 15%) rather than any daily value.** Every other subtopic is flat within Poisson noise on a base of 11–33 records/day; nothing else in this figure is a trend.

Emerging this week. Three genuinely new lines. (i) **Source-resolved exposure** — Dales et al. separate wildfire-sourced from background $PM_{2.5}$ and find +6.07% (5.75–6.39) headache ED visits for the former against a null for the latter; Zhou M et al. attribute 27% of India's population-weighted $PM_{2.5}$ to transboundary transport, more than any domestic sector. (ii) **Ultrafine and number-based metrics** — 6 of 131 records this week use UFP or particle-number exposure (Ramsden, Zhou 2026b, Rusconi, Zhou L, Saeedi, Blanco), against essentially none in the preceding month's dailies. (iii) **Oxidative potential as a modelled quantity** — Mishra et al.'s KM-OP predicts OP from composition at $R^2 > 0.75$, which is the step that makes speciated sensing actionable.

Fading. Nothing has genuinely faded in six days, and claiming otherwise would be noise-fitting. The apparent drop in cardiovascular/metabolic records (4 on 29 July, 2 on 30 July, 0 on 31 July and 1 August) is journal-issue timing on an 11–25-record base, not a decline.

Flat but load-bearing. Exposure assessment & modelling (17 records) and mechanistic toxicology (11) held steady all week and carry the field regardless of the weekly headline. Reproductive/developmental (10) is the cluster with the most consistent effect estimates — Xie's pubertal-timing HRs, Li's stillbirth HRs, Sieber's LBW OR and Pai's incense aORs are all in the same direction with intervals that exclude the null.

Study architecture

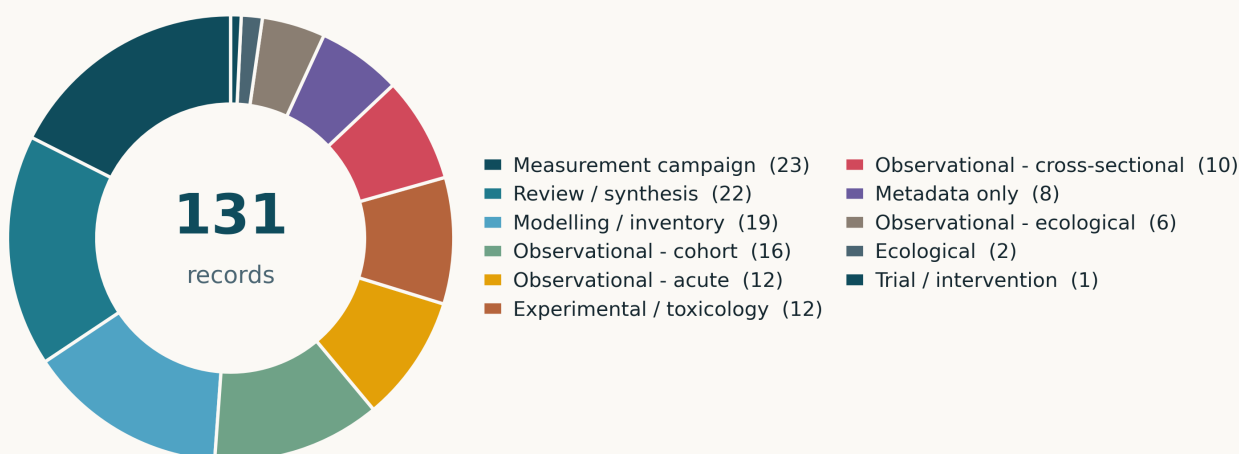


Fig. Pooled study architecture. Measurement campaigns lead at 23, but **reviews and syntheses are effectively tied at 22 — one record in six of this week's corpus is a review**, a high share that reflects a mid-year cluster of review issues rather than new primary evidence. Observational designs together contribute 36 (16 cohort, 12 acute, 6 cross-sectional, 2 ecological), and **exactly one record in 131 is a trial or intervention** — the evidence base this week is almost entirely observational and synthetic.

Pooled analytics

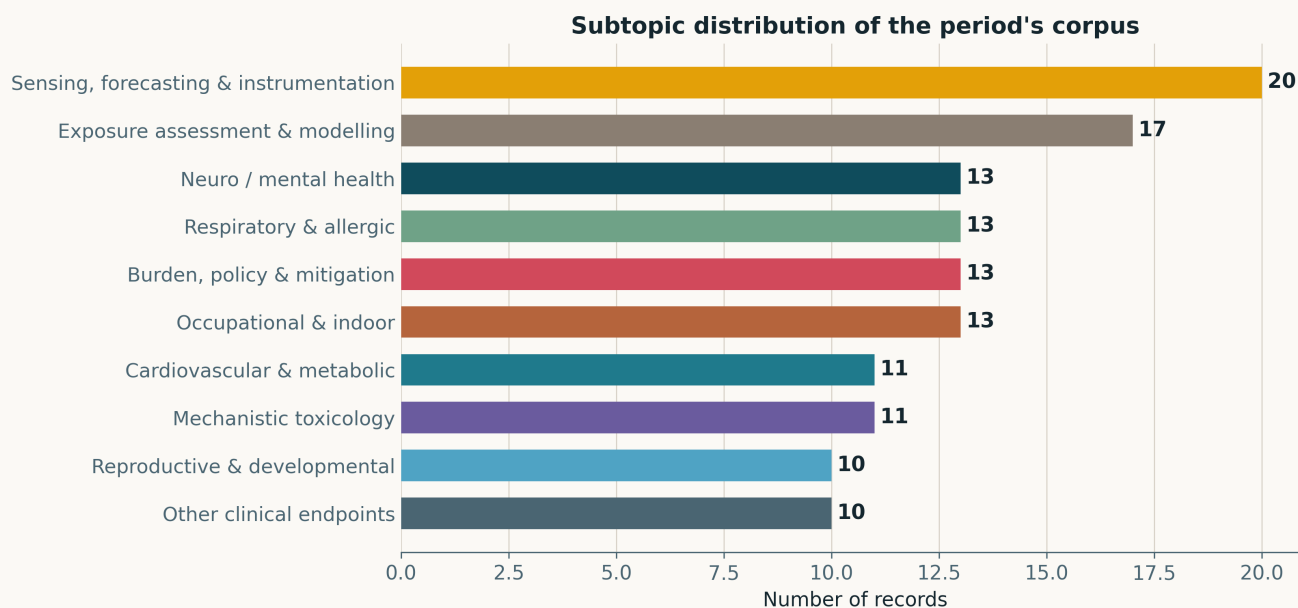


Fig. Pooled subtopic distribution, 27 July 2026 to 1 August 2026. Sensing (20) and exposure assessment (17) together account for **28% of the week** — the methodological half of the field is, for once, better represented than any single health cluster. Neuro/mental health, respiratory/allergic, burden/policy and occupational/indoor tie at 13.

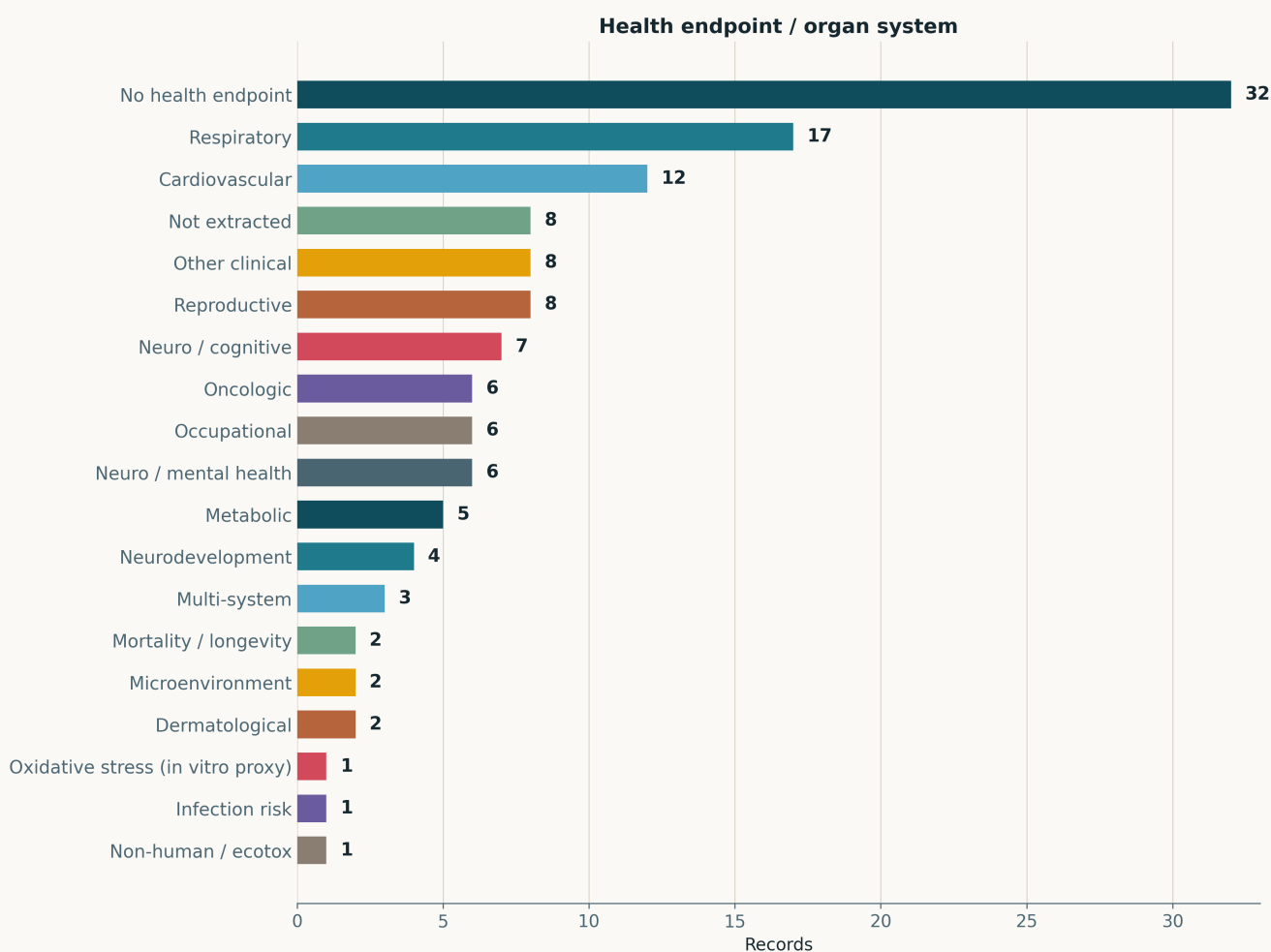


Fig. Pooled health endpoint by organ system. **Thirty-two records carry no human health endpoint** and eight more were surfaced without an abstract — together 31% of the week, which is what a corpus with a healthy instrumentation and source-attribution component looks like. Among records that do carry an endpoint, respiratory (17) and cardiovascular (12) lead; the long tail below 3 records is a single study each and should not be read as a trend.

Quantitative associations reported in the period's corpus (1 of 2)

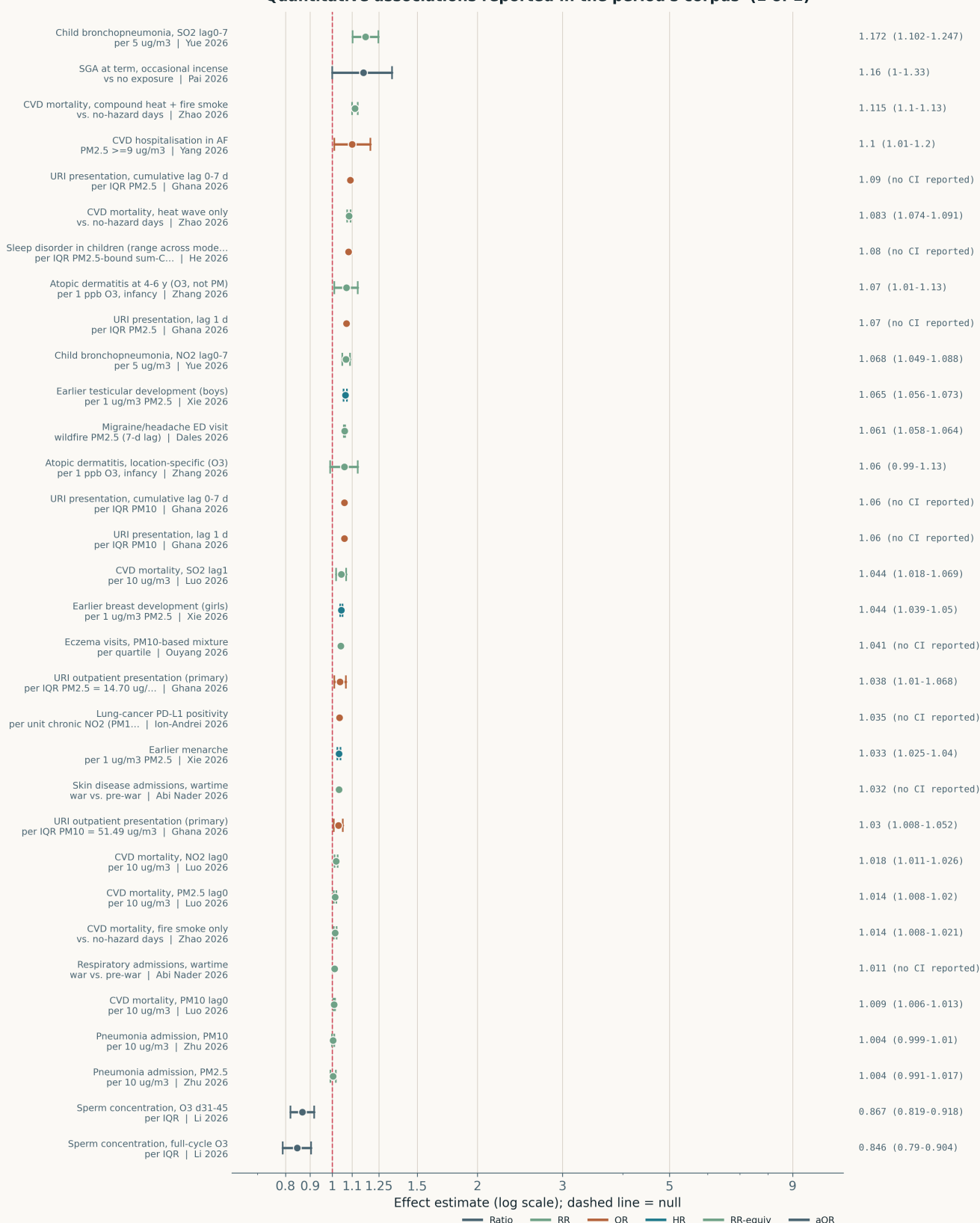


Fig. Pooled effect estimates for the week — read as a caterpillar plot, not a meta-analysis. No summary diamond is drawn and none should be. **Pooling is not defensible here and no I^2 is reported**, for four separate reasons: exposure contrasts differ (per $1 \mu\text{g m}^{-3}$ in Xie, per IQR in Sánchez, a $\geq 9 \mu\text{g m}^{-3}$ threshold in Yang, a percentage-change conversion in Dales); outcomes differ across ten organ systems; several estimates are multiple contrasts from one paper and one cohort (Yang contributes 3, Xie 3, Ghana AIRCLARE 6, Luo 4), so they are not independent; and 7 of the 48 estimates have **no published interval at all** (Mokoena $\times 3$, Ouyang $\times 2$, Abi Nader $\times 2$) and are plotted as points. The one subset that is internally comparable is the four respiratory RRs per $10 \mu\text{g m}^{-3}$ from Zhu and Yue — and within that subset the particulate terms (1.004, 1.004) are null while the gas terms (1.221, 1.068, 1.172) are not.

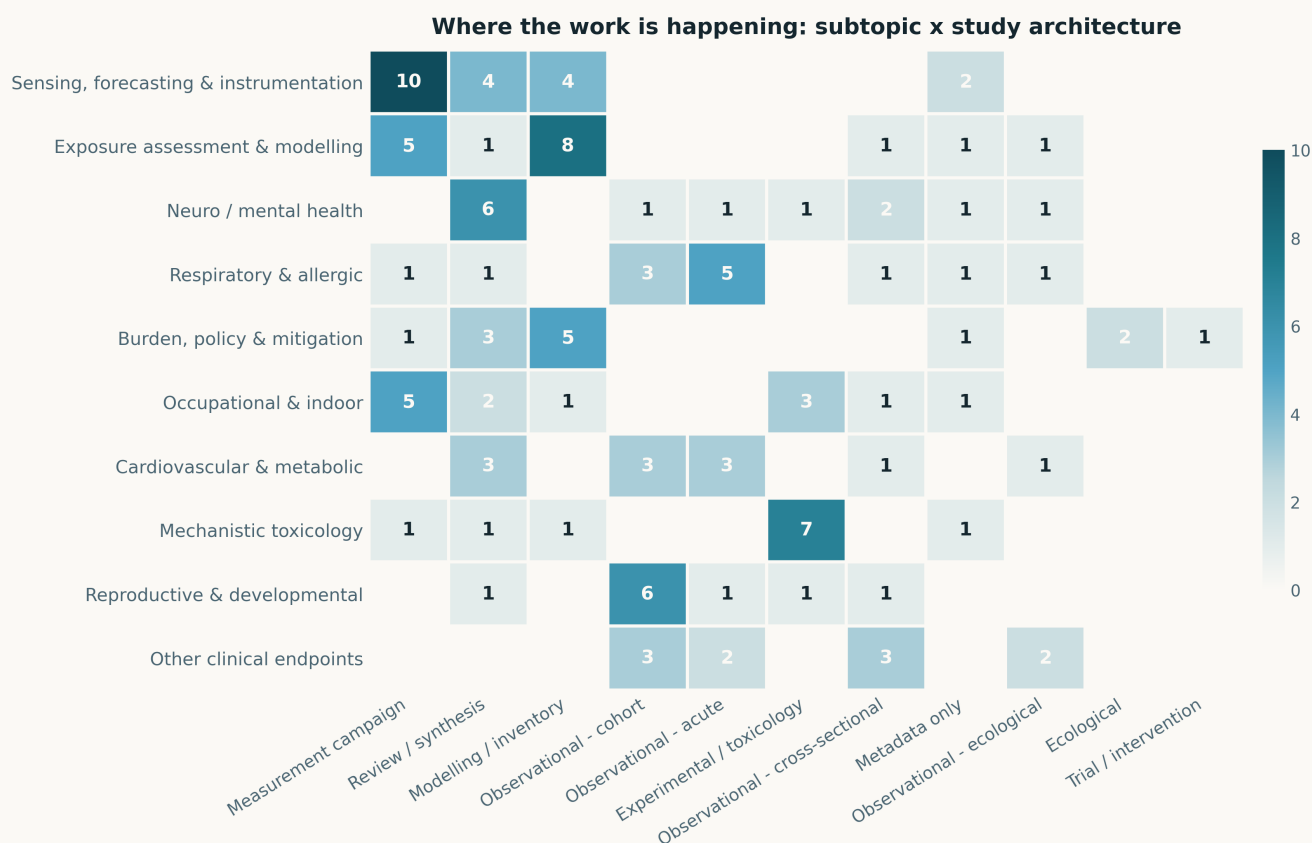


Fig. Pooled subtopic × study-architecture cross-tabulation. **The persistently empty cells are the finding.** Mechanistic toxicology has no cell in any observational column — eleven records, no bridge to human biomarker work. The sensing row has no cell in any cohort or case-crossover column: across 131 records and six days, **no study characterised an instrument and used it to estimate a health effect in the same paper.** Eight records come closest (Wang's SARS-CoV-2 surveillance, Patton's outdoor risk assessment, Sánchez's asthma admissions, Chen's 29-hazard mixture, Blanco's UFP-cognition analysis) but in every case the exposure instrument is inherited, not evaluated.

Groups, venues and most-cited work

Recurring author and group clusters. There are none, and the apparent ones are an artefact worth naming. Surname frequency over the pooled corpus returns Chen (5), Zhou (5), Ding, Park, Lu, Wang, Zhang and Li (3 each) — but these are distinct research groups sharing common surnames, not repeat output. The 1 August issue alone carries two unrelated first-author Zhous (Liyuan Zhou on Bakhoor emissions, Mi Zhou on India source attribution), which is why they are disambiguated as “Zhou L” and “Zhou M” in the register. **First-author surname is not a group identifier and should not be used as one;** a real cluster analysis needs ORCID or affiliation strings, which this pipeline does not currently store. That is a state-schema gap, logged as such. The only genuine programmatic stream visible is the Henan Rural Cohort (Zhang, 1 Aug) and the Chongqing Child Health Cohort (Xie, 27 Jul), each contributing one record.

Venue concentration. Nineteen journals contributed more than one record. *Toxics* leads at 7, *J Environ Sci (China)*, *J Hazard Mater* and *Atmos Environ* at 6 each, *Front Public Health* and *Environ Sci Pollut Res* at 3, *Atmos Chem Phys* at 3. Two of these are structural rather than substantive: the *Atmos Environ* count is entirely the Crossref by-journal leg activated on 1 August and reflects a deposit window, not a research surge; the *Toxics* cluster is a themed-issue effect. **No single-journal cluster this week represents an independent replication.**

Most-cited work of the period. Not reported, because it would be noise. Every record in this corpus entered an index within the last six days; citation counts where available are 0–5 and are dominated by how long a preprint sat before formal publication, not by influence. Ranking a week-old corpus by citations measures publication lag. The metric becomes meaningful in the monthly and yearly rollups and is deferred to them.

Trial registry watch

From `state/trials.json`: registrations created or updated inside the window. Daily issues accumulate; the rollout is where they surface.

New this week — three sham-controlled filtration trials (surfaced 1 August on an intervention sweep; all three are indoor HEPA filtration against a sham unit, which is the correct comparator and rules out the placebo-and-noise objection that sinks open-label purifier studies).

- ▶ **NCT05867381** — *Slowing Atherothrombosis Progression Through Indoor Air Filtration*, University of Southern California. Recruiting, N/A phase, **n = 112**, crossover, HEPA vs sham, adults with ischaemic heart disease history, primary completion **June 2028**. Crossover on a progression endpoint is the right design for a within-person exposure contrast, but 112 participants against an atherothrombosis imaging endpoint is thin: the indoor PM_{2.5} reduction a HEPA unit delivers (~30–60% of indoor mass) implies an effect far smaller than the between-person variance in plaque progression. Expect a surrogate-endpoint result.
- ▶ **NCT05718245** — *Using Indoor Air Filtration to Reduce PM_{2.5} Cardiometabolic Effects in At-risk Individuals*, USC. Recruiting, **n = 52**, HEPA vs sham, primary completion **December 2026** — the nearest-term readout on the list. At **n = 52** this is powered for biomarker change (blood pressure, glucose, inflammatory markers), not for events, and should be read that way.
- ▶ **NCT05016271** — *Health Benefits of Air Purifiers in Primary School Students*, Fudan University. Active, not recruiting, **n = 110**, HEPA-equipped vs non-HEPA ventilation in classrooms, cardiopulmonary function. **Primary completion was June 2022 and no results have been posted** — four years is well past the reporting expectation and is itself worth logging.

Is the endpoint capable of detecting the effect? For all three: not for hard outcomes. A sham-controlled filtration trial changes personal PM_{2.5} by single-digit $\mu\text{g m}^{-3}$ over weeks to months; the cohort literature in this same week (Luo: CVD-mortality RR 1.014 per $10 \mu\text{g m}^{-3}$; Yang: OHCA RR 1.02) implies effect sizes that a 52–112-person trial cannot resolve on events. **These trials are mechanism studies wearing RCT clothing**, and their value is the biomarker and personal-exposure data, not the clinical endpoint.

Standing watch list. **NCT05874479** — *Reducing Air Pollution Exposure to Lower Blood Pressure Among NYC Public Housing Residents* (NYU Langone, **n = 440**, active vs sham portable air cleaner, primary completion September 2027). Logged 29 July; at **n = 440** it is the only trial on the list plausibly powered for a blood-pressure endpoint. Nearest readout to watch: **NCT05718245, December 2026**.

Open gaps

- ▶ **Measurement–epidemiology coupling: still zero.** Across 131 records, no study both characterised an instrument and estimated a health effect with it. Blanco et al. came closest — a mobile UFP campaign feeding a cognition analysis in the same paper — but the monitoring instrument itself is not evaluated, only the campaign design. This is the same gap the 27 July issue named and it has not moved.
- ▶ **Ultrafine and speciated metrics: improving, from nothing.** 6 of 131 records (4.6%) use UFP or particle-number exposure and 3 use speciated composition. That is up from near-zero, entirely because the aerosol-science journals are now being read. It remains badly out of proportion to the deposition physics: the fraction of deposited particle *number* that reaches the alveolar region is dominated by exactly the size range that 95% of this week's records do not measure.
- ▶ **Geographic coverage vs. attributable burden: the gap narrowed.** 13 of 131 records (10%) come from South Asia or sub-Saharan Africa — India 7, Ghana, Uganda, South Africa, Bangladesh and a three-country Gambia/Kenya/Mozambique study. Against China's 42 (32%) this is still inverted relative to attributable burden, but it is the best week on this measure so far, and two of the records (Nyoni, Ghana AIRCLARE) are primary measurement rather than secondary analysis.
- ▶ **Calibration transferability: reported, and it fails.** Three records tested transfer explicitly. Ginsburg et al. (29 Jul) found US-derived PurpleAir corrections do not transfer outside North America across 1,013 globally co-located sensors, with region-specific corrections needed to meet EPA 24h targets. Kumpika et al. report $R^2 = 0.974$ but on single-site, single-season, single-model data — co-located fit, not held-out deployment. Qian et al. report chamber performance only. **No record this week reported held-out deployment performance at a site not used in fitting.**
- ▶ **Negative and null results: present, which is a good sign.** Zhu et al.'s null $PM_{2.5}$ pneumonia RR, Blanco et al.'s null UFP–cognition estimate, Yang et al.'s null $PM \times$ heat interaction and the AFRI-c filtration process evaluation all published as nulls. A corpus with no nulls would be a reporting-bias signal; this one has four in six days, which is consistent with a literature that is at least sometimes publishing what it finds.

Where this is heading — judgement, not measurement

The methodological demand the field is converging on is **source- and size-resolved exposure with a validated error term**. Three separate literatures arrived at it independently this week: epidemiology (Dales's wildfire-vs-background split, Zhu's null mass term), exposure modelling (Saeedi and Blanco on validation and design), and aerosol science (Zhou L on number-dominated indoor emissions, Mishra on predicting oxidative potential from composition). None of them cites the others. That convergence is the strongest signal in the week and I would put moderate-to-high confidence on it persisting.

What would change the picture: a well-powered cohort finding a robust total-mass association after adjusting for source and composition would undercut the whole line, and none of this week's records tested that. The mass-null results here come from single-pollutant models over correlated combustion tracers, which is a weak basis for demoting mass. **I would not yet write " $PM_{2.5}$ mass is the wrong metric" in a thesis introduction** — I would write that its measured association is increasingly sensitive to source attribution and model specification, which is a defensible and much narrower claim.

For a distributed low-cost PM sensing programme the operational reading is specific. First, **number-based channels are now worth the extra cost**: the week produced two independent results (Bakhoor UFP emission rates exceeding sidestream cigarette smoke; 6 UFP records against essentially zero the previous month) that a mass-only network cannot see. Second, **validate on held-out sites, not held-out samples** — Saeedi's 217%→79% collapse is what random-CV validation costs, and it applies directly to any network calibration reported as " R^2 against co-located reference". Third, **instrument ageing and firmware behaviour are first-order error terms**: Kumpika's duty-cycling result and Le's CL61 laser-power/calibration-factor finding both describe degradation that a deployed node cannot self-detect. A co-location rotation schedule is not optional infrastructure.

Confidence that this week was informative: high — but note that the single largest change in the corpus was a fix to *this pipeline*, not to the literature. Nine of the twenty sensing records exist in this rollout because a harvest leg was built on Friday. The literature did not change; the instrument reading it did.

Paper-level digest — the week by subtopic

Every record of the week, grouped by subtopic and ordered by first author within group. **123 of 131 records carry the full critical summary written when they were first screened** — design, effect size with interval, the limitation that threatens the inference, and the line on relevance to distributed sensing. Those summaries are reproduced *verbatim* from the archived daily issues; nothing was re-written for this rollup, and the index is rebuilt by parsing `issues/digest_*.tex` rather than retyped.

The remaining **8 records were surfaced by the Crossref by-journal leg on 1 August and have no abstract** deposited in Crossref, Europe PMC or PubMed. They are listed on title, journal and DOI alone. **No summary is offered for them and the absence of criticism is not approval** — they are unread, and are marked as such.

Sensing, forecasting & instrumentation

20 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (18 OF 20 RECORDS IN THIS SUBTOPIC)

Aher & Ravindran 2026 Environ Sci Pollut Res

Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy

Letter responding to a 2026 forecasting review. The substantive critique is that the review optimises for predictive accuracy while under-treating what actually breaks operational systems: model transferability into data-sparse regions, uncertainty quantification, interpretability of deep architectures, and input data-quality constraints. It points to adaptive mesh refinement, AI-assisted chemical transport models and hybrid physics–ML frameworks as the productive direction. No new results — but the transferability-into-data-sparse-regions argument is precisely the problem low-cost sensor networks exist to solve, and the letter states it cleanly.

doi: 10.1007/s11356-026-38087-y

Ainembabazi et al. 2026 Toxics

Trends in atmospheric pollution research in Uganda, 1990–2025: a scoping review

Structured scoping review across Google Scholar, Web of Science and PubMed. Output was negligible before 2010, expanded after 2015 and surged 2020–2025. PM_{2.5}/PM₁₀ dominate throughout while NO₂, SO₂, CO and O₃ are underrepresented; studies cluster in urban Kampala with almost no rural ambient monitoring; and the methodological trajectory runs from proxy and gravimetric approaches to **low-cost sensors, portable monitors and satellite-derived products**. The authors' own conclusion is the operative one: short-term, spatially constrained campaigns predominate, leaving no long-term or nationally representative record. This is the clearest statement in today's corpus of the niche a sustained calibrated low-cost network fills — and a reminder that Google Scholar inclusion makes the sampling frame non-reproducible, so the counts should be treated as indicative.

doi: 10.3390/toxics14070542

Feng et al. 2026 Meas Sci Technol

Calibration strategies and measurement frameworks for low-cost PM sensors

Review spanning ambient, mobile, indoor and workplace deployments (community networks, vehicle-mounted platforms, smart-manufacturing floors, mining). Compares purely data-driven, purely physics-based and hybrid physics-informed ML calibration on accuracy, transferability, compute cost and interpretability: non-linear models reach sub-10 $\mu\text{g m}^{-3}$ error but resist embedded deployment and depend on representative training data; **no strategy is universally optimal**. Names the persistent barriers plainly — sensor fouling, data governance, model generalisation, hardware limits — and points to self-cleaning optics, federated learning and knowledge-distilled on-device calibration. Narrative rather than quantitative synthesis, with no pooled performance estimate, so it frames rather than settles the question.

doi: 10.1088/1361-6501/ae6298

Ginsburg et al. 2026 Aerosol Sci Technol

Global and regional calibration of low-cost PM_{2.5} sensors (PurpleAir)

1,013 PurpleAir units co-located within 500 m of reference monitors on six continents, Jan 2023–Mar 2025, assessed at 1-h and 24-h averaging by R^2 , RMSE, nRMSE, precision (CV) and AQI category correctness. Raw readings overestimate PM_{2.5} in most regions and miss EPA performance targets; US-derived corrections show limited transferability outside North America; a data-driven global quadratic helps in Australia/NZ and hurts elsewhere; **regional equations meet EPA 24-h thresholds in multiple regions and lift 24-h AQI correctness to $\geq 77\%$** . Limitation: co-location proximity (≤ 500 m) is a proxy for comparability, so some regional “bias” is spatial-gradient error, and continents are represented very unevenly. Still the strongest available empirical case that LCS network calibration must be local.

doi: 10.1080/02786826.2026.2676293

Huda et al. 2026 J Air Waste Manag Assoc

Low-cost sensor network for wildfire monitoring at remote community locations

Solar-powered microsystem (PM_{2.5}, CO, CO₂, O₃, tVOC) with a microcontroller scheduler and mesh radio, built for forest deployment in northwestern Alberta where there is no cellular or Wi-Fi coverage. Reported PM_{2.5} RMSE $< 5 \mu\text{g m}^{-3}$; mesh range validated at 1–5 km through urban structures and vegetated terrain; 50×37 cm panel sized for treetop mounting with energy-aware duty cycling. This is an engineering report, not an evaluation: no co-location period, drift characterisation or seasonal uptime statistics are given, and winter operation is described as prospective. Useful for the power and comms architecture rather than the data-quality claim.

doi: 10.1080/10962247.2026.2665375

Kumpika et al. 2026 Environ Pollut Bioavailab

Correction of humidity and sensor ageing effects in low-cost PM_{2.5} light-scattering sensors

Field campaign in Chiang Mai with PMS7003 units across a wide relative-humidity range, validated against a government BAM and cross-checked with dry-air measurement through a diffusion dryer. Polynomial corrections up to third order were fitted; **agreement with BAM reached $R^2 = 0.9737$** after correction, with the largest gains at elevated RH. The second result is the more interesting one: **restricting operation to 20 min/day measurably reduced long-term drift** relative to continuous running, implicating cumulative optical-chamber exposure rather than elapsed calendar time as the ageing mechanism. Single-site, single sensor model, single season, so the polynomial coefficients are not transferable — but duty-cycling is a design lever, not a calibration constant, and it applies to any continuously powered node.

doi: 10.1080/26395940.2026.2616098

Le et al. 2026 Atmos Meas Tech

Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling

Three years of operational data from multiple CL61 units — a depolarisation-capable elastic backscatter lidar — assessed for inter-unit consistency, drift and bias. Most early production units showed **progressive laser-power decline with rising background noise**, and manual calibration against atmospheric targets at Lindenberg showed that **once laser power falls below 40% the internal calibration factor stops compensating**. Instrumental background and bias vary with temperature; a correction and uncertainty-estimation method is given. Under low aerosol load the bias correction shifts volume linear depolarisation ratio by up to 0.005, while the volume-to-particle depolarisation difference reaches 0.1 — so molecular correction, not instrumental bias, dominates quantitative aerosol interpretation at 910.55 nm. One unit's signal loss was traced to window fogging from insufficient firmware-driven heating. Instructive as a template: silent, firmware-mediated degradation is exactly the failure mode a distributed network cannot detect without co-location.

doi: [10.5194/amt-19-4923-2026](https://doi.org/10.5194/amt-19-4923-2026)

Mohapatra et al. 2026 IEEE Access

IoT-machine-learning proof-of-concept framework for real-time urban particulate matter prediction

PMS7003 on an Arduino module at campus scale, with outlier treatment, normalisation and temporal indexing feeding random-forest and polynomial regression, plus k -means clustering to identify persistent high-exposure zones. Predictive performance **exceeds 85% for $PM_{1.0}$ and $PM_{2.5}$ but degrades sharply at 5 and 10 μm** , which the authors correctly attribute to discretised, low-variability sensor output rather than to the model — a useful negative result about the coarse channels of optical counters generally. No reference co-location, single site, and the models predict the sensor's own future readings rather than true concentration, so "performance" here is self-consistency. Included for the size-channel finding only.

doi: [10.1109/access.2026.3651996](https://doi.org/10.1109/access.2026.3651996)

Msigwa et al. 2026 IEEE Internet Things J

AIoT real-time sensing and calibration of low-cost PM sensors (TCM-Net)

Customised PMS7003 units with externally PWM-controlled fans to stabilise airflow, on a oneM2M-compliant IoT platform, calibrated by a transformer + temporal-convolutional network with a memory-aware module and trained against a co-located certified reference. Reported RMSE $0.325 \mu g m^{-3}$, R^2 0.999. **Those figures sit below the reference instrument's own measurement uncertainty** and cannot be a valid generalisation estimate; the near-certain explanation is temporally adjacent train/test splits on autocorrelated series rather than held-out deployment. The forced-airflow hardware idea is worth taking; the performance number is not.

doi: [10.1109/jiot.2025.3631631](https://doi.org/10.1109/jiot.2025.3631631)

Nurseitov et al. 2026 Information

Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter

Six custom nodes (Winsen ZPHS01B on Raspberry Pi Zero 2 W, ~US\$100/node) deployed across Almaty over a February–April campaign, **>70,000 measurements**, with three engineering contributions: an active airflow-stabilisation enclosure decoupling sampling from wind, context-aware adaptive edge filtering cutting transmitted volume ~40%, and a segmented telemetry pipeline. Cross-validation was against a consumer monitor with documented traceability rather than a reference analyser ($R^2 = 0.89$ – 0.95), and the authors state explicitly that the network provides *internally consistent* rather than reference-equivalent concentrations — unusually honest framing. **Daily mean $PM_{2.5}$ exceeded the WHO 24 h guideline on 84% of days** (February 54.4 vs March $21.9 \mu g m^{-3}$); a $PM_{2.5}/PM_{10}$ ratio near 0.96 plus higher weekend levels points to residential coal heating. That ratio is high enough to warrant suspicion of the coarse channel rather than confidence in the source inference.

doi: [10.3390/info17070642](https://doi.org/10.3390/info17070642)

Patel et al. 2026 Proc AAAI

Scalable air-quality sensor placement via gradient-based mutual information maximization

Treats sensor coordinates as differentiable parameters and maximises mutual information by continuous optimisation, so **runtime is independent of both the number of placements and candidate-grid density** — the two terms that make greedy MI intractable at national scale. Beats random placement and the maximum-predictive-variance heuristic on a continental $PM_{2.5}$ dataset and approaches greedy MI regionally, orders of magnitude faster. Caveat: MI is computed under an assumed GP covariance, so placements inherit that model's stationarity assumptions, which is exactly where real PM fields misbehave near sources. Immediately useful for network-expansion planning under a fixed sensor budget.

doi: [10.1609/aaai.v40i46.41257](https://doi.org/10.1609/aaai.v40i46.41257)

Patton et al. 2026 Environ Sci Pollut Res

Human health risk of particle exposure in outdoor environments, southwestern Pennsylvania

ROCIS low-cost monitoring across 290 homes in Allegheny County, 2016–2021; optical particle counts $>0.5 \mu m$ converted to $PM_{2.5}$ -equivalent mass, census tracts classified by environmental-justice status and industrial proximity, GEE models plus non-cancer hazard quotients. **33.8% of sites exceeded the WHO 24-h $15 \mu g m^{-3}$ guideline and 66.6% the EPA annual $9 \mu g m^{-3}$ reference** on 3-week site medians; winter concentrations significantly elevated. The count-to-mass conversion is the weak link — it assumes a fixed density and size distribution that industrial plumes violate — and the authors correctly flag the Sewickley extreme (median $19.3 \mu g m^{-3}$) as resting on three homes. Directly relevant as a template for community-scale LCS deployment with explicit EJ stratification.

doi: [10.1007/s11356-026-38083-2](https://doi.org/10.1007/s11356-026-38083-2)

Qian et al. 2026 Atmos Meas Tech

Enhancing accuracy of indoor air-quality sensors via automated machine-learning calibration

Multi-stage AutoML calibration chaining field sensors to intermediate drift-correction reference sensors to a reference-grade instrument, with **separate models fitted below and above a concentration break** for clean air versus pollution events. Evaluated in a controlled indoor chamber with two sensor models against diverse real indoor sources under uncontrolled ambient conditions: $R^2 > 0.90$, **NRMSE and sMAPE roughly halved** against uncalibrated output, bias effectively removed. The split-range design is the transferable idea — a single global calibration is being asked to serve two distinct optical regimes and it fails at both ends. Chamber conditions make the result an upper bound on field performance, and no ageing term is included, which the Kumpika result suggests is the dominant long-run error.

doi: [10.5194/amt-19-603-2026](https://doi.org/10.5194/amt-19-603-2026)

Reis 2026 Sensors

Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference

Architecture and system-level optimisation of a multi-parameter node coupling a low-power microcontroller to a neural inference accelerator, with adaptive duty-cycling, hierarchical power domains and quantised fixed-point models doing three-class normal/anomalous/critical classification at the edge. The reported figures — **94% inference accuracy, 0.87 ms latency, $\approx 2.9 mWh$ average power, $\approx 88\%$ reduction in LoRaWAN transmissions** — come from MATLAB/Simulink simulation, not deployment, and there is no co-location against a reference instrument anywhere in the study. Treat the energy budget as a design target and the accuracy figure as unvalidated. The duty-cycling and hierarchical-power argument is the reusable part, and it is also the part that does not depend on the classifier working.

doi: [10.3390/s26020703](https://doi.org/10.3390/s26020703)

Shi et al. 2026 J Environ Manage

Adaptive Spatiotemporal Graph Transformer for multi-site multi-step PM_{2.5} forecasting

AST-GT with five components: adaptive non-stationary normalisation for time-varying distribution drift; Transformer temporal encoding; multi-source context fusion (geography, meteorology, co-pollutants); graph-based dual-scale spatial attention; and a temporal-spatial cooperative attention block. Validated on Beijing and India multi-site datasets. The non-stationarity handling and the explicit treatment of *uneven site distribution* are the parts worth reading if you work with sparse monitoring networks.

doi: 10.1016/j.jenvman.2026.130515

Silva et al. 2026 J Air Waste Manag Assoc

Low-cost sensors for indoor air-quality monitoring: systematic review of accuracy, applications and limitations

PRISMA review, **24 studies retained from 2,736 screened**, covering optical particle counters and NDIR CO₂ against reference-grade instruments in real settings. Strong correlations are achievable under controlled or semi-controlled conditions, but performance degrades with humidity, temperature and source type, and the authors flag methodological heterogeneity, near-absent inter-device validation, regional bias and **un-addressed sensor ageing and cross-sensitivity** as systematic gaps. The operational conclusion — that most devices do not meet regulatory compliance requirements and standardised validation protocols do not exist — is well supported and worth citing directly. As a review it inherits the heterogeneity it documents, and a 24-study base is thin for the breadth of the claims.

doi: 10.1080/10962247.2026.2624795

Tsuchiya et al. 2026 Biosensors

Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices

Real-time bioaerosol sensor across 10 office spaces in four Japanese regions, summer and winter. Indoor FAP generally <100 particles L⁻¹, peaking at 195 p/L in winter; several buildings higher in winter, attributed to RH<40% promoting resuspension. Size-resolved FAP fraction rises sharply with diameter (40–74% at 2–5 μm; 96–100% >5 μm). Directly relevant to anyone deploying optical/fluorescence sensors — the coarse-mode dominance constrains what a fluorescence channel can be interpreted to mean.

doi: 10.3390/bios16070380

Wang et al. 2026 Viruses

Joint monitoring and early warning of SARS-CoV-2 in outdoor PM_{2.5} and wastewater

Twelve months (2023) of paired outdoor PM_{2.5} filter and wastewater sampling in Fuzhou, RT-qPCR for SARS-CoV-2 RNA, associations with pollutants and meteorology by GAM and DLNM. **45.9% of PM_{2.5} samples were positive**; viral load in PM_{2.5} rose with CO, PM_{2.5} mass and atmospheric pressure and fell with temperature and sunshine duration, with the **strongest lead over outpatient COVID cases at ≈ 2 days** against lag 0 for wastewater. The direct interest for a distributed-sensing group is that a routine gravimetric filter stream doubles as a bioaerosol surveillance substrate with a two-day lead — but only if filters are archived and chain-of-custody preserved, which most low-cost optical nodes cannot do at all. The weakness is that RNA detection on a filter conflates aerosol transmission with resuspended or deposited material, and no infectivity assay was attempted; the “early warning” claim rests on ecological time-series correlation with outpatient counts.

doi: 10.3390/v18070788

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (2 RECORDS)

Park et al. 2026 Atmos Environ

Diagnosing explainability of machine learning: transformer-based PM_{2.5} estimation in Seoul

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122264

Zheng et al. 2026 Atmos Environ

Augmented transformer framework for PM_{2.5} chemical composition via one-step LDI single-particle aerosol mass spectrometry

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122260

Exposure assessment & modelling

17 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (16 OF 17 RECORDS IN THIS SUBTOPIC)

Blanco et al. 2026 J Expo Sci Environ Epidemiol

Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences

A reference roadside UFP model built from repeated measurements at 309 locations was used with the Adult Changes in Thought cohort ($n = 5,283$) to test late-life cognitive function, then **600 km of on-road measurements were subsampled** to emulate alternative campaign designs varying visit frequency, spatial balancing and sampling time, with universal kriging plus partial least squares. The reference model gave a null: **CASI-IRT +0.007 (95% CI −0.013 to 0.027) per 1,900 pt/cm³**. Alternative designs reproduced the null but attenuated it; route-based sampling and very short 4-visit campaigns produced the most variable estimates, while temporal and plume adjustments changed little. Testing design sensitivity against a null limits what can be concluded — attenuation toward zero is unobservable when the reference is zero — and the authors say so. The design guidance, that visit count and out-of-hours coverage matter more than post-hoc adjustment, is what carries.

doi: 10.1038/s41370-026-00845-y

Chen et al. 2026 J Hazard Mater

Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities

615 daily PM_{2.5} samples across Tianjin, Qingdao, Shanghai, Xiamen and Guangzhou during the cold-season transition, with nitrate $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ fed to a Bayesian isotope mixing model against an emissions-inventory prior. Shipping-derived nitrate ranged **17.6±4.7% (Tianjin) to 37.9±11.4% (Xiamen)**. Northern cities show shipping contribution rising with pollution severity and marine air-mass association; southern cities show it falling as inland transport intensifies. The inference threat is the mixing-model prior — source endmember $\delta^{15}\text{N}$ ranges for shipping overlap substantially with those for coal combustion, and no endmember sensitivity analysis is reported. Relevant to sensing because it quantifies how much of coastal PM_{2.5} mass a mass-only monitor cannot attribute.

doi: 10.1016/j.jhazmat.2026.143129

Chen et al. 2026 J Hazard Mater

Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability

Exposome-wide association study over 29 residential hazards — air pollutants, industrial VOC emissions, highway density and proximity to 13 pollution-source classes — in 7,092 North Carolina adults (2013–2022), stratified at the median census-tract Social Vulnerability Index, with quantile g-computation for the mixture. In the overall population only proximity to dry-cleaning-solvent-contaminated sites was associated with

asthma. In high-SVI tracts, **eight hazards — predominantly PM_{2.5} constituents and VOCs — were positively associated at FDR < 0.05**, and each one-quintile increase in the hazard mixture carried **35% higher asthma odds (95% CI 6–73%)**. Toluene, dry-cleaning proximity, PM organic matter and carbon monoxide each contributed over 10% of the joint association. Hazard correlations were themselves stronger in high-SVI areas, which is both the substantive finding and the analytical problem: with collinearity differing by stratum, the g-computation weights are not directly comparable across strata. Cross-sectional, self-reported physician-diagnosed asthma, residence-only exposure.

doi: 10.1016/j.jhazmat.2026.143083

Chengdu supersite 2026 J Environ Sci (China)

Gas-phase reactions accelerate nitrate formation in spring and early summer, Sichuan Basin

Urban supersite speciation of PM_{2.5} over April–June 2018–2021 in Chengdu, targeting the under-studied warm-season nitrate problem. During complex PM_{2.5}/O₃ episodes nitrate rose **47.8%, from 5.52 to 8.16 $\mu\text{g m}^{-3}$** , with its share of PM_{2.5} climbing from 13.4% to 22.2%. The NO₃[−]/PM_{2.5} ratio scales with aerosol liquid water, and rapid early-morning ALW growth drives hygroscopic growth and heterogeneous production on top of the gas-phase channel. Directly consequential for optical sensing: a nitrate- and water-dominated aerosol has a very different hygroscopic growth factor and refractive index from the winter carbonaceous mixture most low-cost sensor calibrations are fitted to, so a humidity correction trained in winter will bias warm-season estimates. Observational and single-city, so the mechanism is inferred from covariation rather than tested.

doi: 10.1016/j.jes.2026.02.002

Ding et al. 2026 Geo-spat Inf Sci

Urban–rural balanced estimation framework for PM_{2.5} exposure across China

Targets a structural bias rather than a model-fit problem: because ground stations cluster in cities, AOD-to-PM_{2.5} retrievals systematically overestimate rural concentrations. Integrates AOD, reanalysis fields and ground measurements under data-assimilation principles, adaptively constraining non-urban estimates with low-weighted reanalysis samples, then ensemble-averages across dynamically weighted background/observation settings. China 2015–2021: **clean-site RMSE 18.18 → 8.31 $\mu\text{g m}^{-3}$** , **national mean PM_{2.5} down ~55%**, with corresponding correction of PM-attributable mortality in low-density areas. The 55% figure is the paper's own counterfactual, validated at clean sites that are still station locations — true rural accuracy remains unvalidated by construction. If even partly right, it implies an inherited upward exposure bias across the satellite-based exposure–response literature.

doi: 10.1080/10095020.2026.2624335

Hu et al. 2026 Geo-spat Inf Sci

Fusion of ground monitoring, satellite retrieval and CTM for seamless hourly 1-km PM_{2.5}

Fills satellite AOD gaps with continuous WRF-Chem AOD, injects physics-informed features (PM_{2.5} temporal components, AOD vertical structure) from the CTM, then predicts with a geospatially aware LightGBM that exploits monitoring-site spatial structure. Site-based cross-validation R^2 0.87, RMSE 13.42 $\mu\text{g m}^{-3}$, MAE 8.68 $\mu\text{g m}^{-3}$, with day-and-night coverage demonstrated on one pollution episode. Site-based CV overstates performance away from stations — the standard weakness — and an RMSE of 13.4 is large relative to the WHO annual guideline, so this is useful for episode mapping and weak for long-term exposure assignment. The physics-feature injection is the transferable step.

doi: 10.1080/10095020.2026.2680855

Nguyen et al. 2026 Eng Res Express

Satellite-driven PM_{2.5} prediction: machine learning and explainable AI

Systematic review of 2015–2025 work integrating satellite products (LEO and geostationary AOD, top-of-atmosphere reflectance, Ångström exponent) with ML/DL. Finds Random Forest still a robust benchmark, hybrid and transformer architectures better at spatiotemporal dependence, and AOD gap-filling plus normalisation the dominant determinants of model stability; positions XAI as the route to physical plausibility. No pooled quantitative comparison and no protocol registration, so the ranking claims are narrative. Adds little beyond Hu and Ding above; retained as a citation anchor for the physics-guided-learning argument.

doi: 10.1088/2631-8695/ae7028

Nyoni et al. 2026 Air Qual Atmos Health

Performance of road-traffic-based exposure proxies against personal PM_{2.5} measurements in three sub-Saharan African countries

One year of filter-based outdoor personal PM_{2.5} from 343 postpartum participants in The Gambia, Kenya and Mozambique, used to validate population-weighted road network density (wRND), Euclidean distance to highway (EH) and to main road (EM). Spearman correlations were weak and **sign-inconsistent across countries**: wRND significant only in Mozambique ($\rho = 0.351$), EH positive in The Gambia (0.335) and Mozambique (0.292) but **negative in Kenya** (-0.224 , $p = 0.018$). Single-proxy random forests gave $R^2 < 0.45$ everywhere; combined models reached $R^2 = 0.52$ (RMSE 31.7 $\mu\text{g m}^{-3}$) in Kenya and 0.60 (8.9 $\mu\text{g m}^{-3}$) in Mozambique. The sampled population — postpartum women — is not representative of general mobility patterns, and only outdoor-filtered personal exposure was measured, so the household cooking contribution that most plausibly drives the proxy failure is inferred rather than partitioned. Still the strongest available empirical case that road-geometry proxies require local validation before use, and a direct argument for node deployment over GIS surrogacy.

doi: 10.1007/s11869-026-02060-y

Pajak et al. 2026 Toxics

Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk

Source-to-receptor assessment of a large landfill tire fire coupling HYSPLIT dispersion modelling against air-quality monitor data with soil sampling on the fireground and in surrounding allotment gardens, screened for Zn as a tire tracer and 16 priority PAHs. Emissions **PM₁₀ 1.34 t and PAHs 17.7 kg**; fireground soil ΣPAH 148.9 mg kg^{−1}; cumulative child ILCR **9.7×10^{-4}** , an order of magnitude above the 10^{-4} regulatory benchmark, with dermal contact rather than inhalation the dominant pathway and elevated risk (10^{-5} – 10^{-4}) persisting at distal residential sites. Ecological risk quotients exceeded unity for PAHs everywhere and for Zn/Cu near the fire. Limitation: no pre-fire soil baseline, so urban background PAH — itself substantial in a coal-heating region — is attributed to the fire; and HYSPLIT at this scale is validated against a sparse monitor network. The transferable point is that acute source events deposit a multi-year soil legacy that airborne monitoring alone will not detect.

doi: 10.3390/toxics14070543

Patwary 2026 Res Sq (preprint)

Spatio-temporal assessment of PM₁₀ and meteorological correlates in Cumilla, Bangladesh

Single Department of Environment monitoring station, January–April 2026, Pearson correlation and trend analysis against temperature, RH and rainfall. Dry-season peak **210.7 $\mu\text{g m}^{-3}$ monthly mean PM₁₀ in February**, roughly 4.7× the WHO 24-h guideline. No instrument model, calibration history or data-completeness statistics are reported, and one station cannot support the “spatio-temporal” framing in the title. Logged because South Asian monitoring records are structurally scarce in this corpus, not because the analysis is sound.

doi: 10.21203/rs.3.rs-10519411/v1

Raj & Dhada 2026 Res Sq (preprint)

Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh

Five Himalayan towns (Solan, Baddi, Shimla, Kullu, Manali) contrasted by land use, 25 June–24 July 2026, 150 site-days, using AQI plus PM_{2.5}, PM₁₀, CO, SO₂, NO₂ and O₃ taken from the public AQI.in platform. The land-use contrast (industrial Baddi vs. tourist hill towns) is the design's one genuine idea. The fatal weakness is provenance: AQI.in aggregates heterogeneous and partly low-cost sources with no published calibration chain, so between-town differences cannot be separated from between-instrument differences. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-10492149/v1

Saeedi et al. 2026 Environ Sci Technol

Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles

Toronto mobile UFP campaign used to compare LUR models trained under random, spatial, temporal and spatiotemporal cross-validation, with and without forward feature selection, each evaluated on a hold-out test set and against *independent* stationary backyard measurements. **Average percentage error fell from ~217% under random CV to ~79% under spatiotemporal CV with FFS.** Random-CV models overfitted, were sensitive to outliers, produced spatial artefacts and exaggerated variable effects, and failed on independent samples — while reporting good internal validation statistics. Spatial folds were designed at scales reflecting UFP autocorrelation, which is the methodological core. The blunt implication: a large share of published LUR performance numbers are measuring data reproduction, and any exposure surface fed into an epidemiological model should carry a spatiotemporally validated error, not a random-CV one.

doi: 10.1021/acs.est.5c12601

Sanchez et al. 2026 Indian J Occup Environ Med

Morbidity related to the level-two eruption of Popocatepetl

Ecological correlational study in communities 15 km from the crater, May 2023, with daily PM₁₀/PM_{2.5} and Kriging used to interpolate ash dispersion because the monitoring network is too sparse to resolve it. 14% of presentations were ash-related, predominantly non-infectious respiratory; Spearman correlations between ash exposure and morbidity were not statistically significant. The authors read this as possible biological adaptation, which the design cannot support — a null in a small ecological study with interpolated exposure is far more likely to be measurement error and low power. Its real value here is as a candid worked example of exposure assignment where no formal network exists.

doi: 10.4103/ijoem.ijoem_146_24

Solórzano-Araque et al. 2026 GeoHealth

Multiscale exposure modelling with LOTOS-EUROS, mobility surveys and particle toxicity

Medellín / Aburrá Valley, 2019. LOTOS-EUROS CTM driven by WRF at 1 km × 1 km; mobility from an origin–destination survey of >180,000 trips across 240 traffic analysis zones; morpho-chemical characterisation of metals and carbonaceous fractions; and a weighted exposure model combining exposure time, concentration and cytogenotoxic indicators. Cluster analysis identified central and southern zones and highway corridors — ~60% of the metropolitan population — exceeding 40 µg/m³ at peak hours. The toxicity-weighting step is the transferable idea.

doi: 10.1029/2025GH001621

Zhang et al. 2026 npj Clim Atmos Sci

High-resolution aerosol liquid water content across the contiguous US using machine learning

ML surrogates trained on GEOS-Chem output to learn ALWC—predictor thermodynamics, then applied to bias-corrected high-resolution inputs, giving **1 km daily ALWC for 2000–2019**. Outperforms the parent CTM on daily variability and spatial heterogeneity. Regional ordering Midwest > East > West is driven by PM_{2.5} mass, composition, temperature and RH; the secular decline tracks sulfate. Demonstrated as a constraint on water-soluble iron variability. The limitation is structural: the ML model inherits every bias of GEOS-Chem's thermodynamic module, so this is a downscaling of a simulation, not an observational product — there is no independent ALWC validation set. Still the most directly useful record here for hygroscopic-growth correction of optical sensors.

doi: 10.1038/s41612-026-01371-2

Zhou M et al. 2026 Atmos Chem Phys

Surface PM_{2.5} in 2022 India: emission updates, WRF-Chem evaluation and source attribution

First WRF-Chem evaluation and source attribution for India for 2022, with an updated residential inventory reflecting the household clean-fuel transition, a plant-level coal-power inventory, a revised secondary-organic-aerosol scheme and improved near-surface mixing. **Annual PM_{2.5} bias 0.2 ± 16.9 µg/m³ (0 ± 31%) across 288 South Asian monitoring sites**; simulated population-weighted annual mean **47.4 µg/m³**. Attribution: transboundary transport 27% (12.8 µg/m³) — larger than any single domestic sector — industry 18% (8.6 µg/m³), residential 15% (7.3, still dominant in the Indo-Gangetic Plain), power 13% (6.1, concentrated in central India). A ±31% annual bias is respectable for a CTM but is the same order as the sectoral shares being separated, so the ranking of industry above residential should be read as provisional. The residential demotion is nonetheless a policy-relevant result about a transition that actually happened.

doi: 10.5194/acp-26-10533-2026

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Delgado Peralta et al. 2026 Atmos Environ

Improving PM_{2.5} simulations in metropolitan Sao Paulo via PMF-based emission inventory adjustment

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122249

Cardiovascular & metabolic

11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Belay et al. 2026 Eur J Prev Cardiol

Long-term bushfire-related PM_{2.5} and coronary plaque burden

365 asymptomatic adults aged 40–70 with a family history of CAD, serial coronary CTA at baseline and 3 years, with daily total and bushfire-attributed PM_{2.5} from validated 5 km models assigned by residential postcode. Per 1 µg m⁻³ cumulative bushfire PM_{2.5}: **total plaque volume +10.14 mm³ (1.45–18.82)** at baseline and +4.92 mm³ (1.33–8.52) at follow-up; total PM_{2.5} associations were weaker. Note the exposure setting — 10-year means of 5.88 and 5.93 µg m⁻³, i.e. near the WHO guideline — which makes a plaque-volume signal notable but also means the bushfire and total metrics are highly collinear at these levels. Postcode-level assignment in a 365-person sample is the binding limitation.

doi: 10.1093/eurjpc/zwag399

Chen et al. 2026 Environ Pollut

Long-term PM exposure and coronary inflammation by pericoronary fat attenuation index

5,528 CCTA patients with a 656-person serial subcohort; FAI extracted for LAD, LCx and RCA by a deep-learning pipeline; individual PM_{2.5}, PM₁₀ and black carbon from a validated 1 km spatiotemporal model; GAM and linear mixed-effects models. Per IQR of 1-year exposure, **RCA FAI rose 7.57 (PM_{2.5}), 8.72 (PM₁₀) and 2.35 HU (BC)**, with accelerated annual FAI progression longitudinally. FAI is technically sensitive to tube voltage, contrast timing and epicardial adiposity, and a retrospective clinical population undergoing CCTA is selected on suspected disease — so absolute HU shifts should not be read as population effects. The local-inflammation endpoint nonetheless complements Belay's structural one.

doi: 10.1016/j.envpol.2026.128840

Chrysohoou et al. 2026 Eur J Prev Cardiol

Preventing cardiovascular disease within the built and living environment: an expert opinion

Expert opinion placing air and noise pollution alongside food environment, green space, physical activity and psychosocial stress in a single built/living/social framework, with inflammation, oxidative stress, neuroendocrine dysregulation and chronic stress as the shared mechanistic substrate. No new data and no quantitative synthesis; its value is as a citable framing document for grant and introduction text, nothing more.

doi: 10.1093/eurjpc/zwag387

Huang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

PM_{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification

10,788 children aged 8–15 across 103 surveillance counties; 3-year mean PM_{2.5} 30.36±10.62 µg/m³; dyslipidaemia prevalence 21.84%. **OR 1.501 (1.028–2.192)**, with no significant association among children with greater vegetable intake diversity. Effect-modification-by-diet claims of this type are fragile to residual confounding, but the sample size is substantial.

doi: 10.3760/cma.j.cn112338-20251120-00843

Kaźmierska et al. 2026 Przegl Epidemiol

Air pollution and cardiovascular disease: review of current data

Standard mechanistic review (oxidative stress, inflammation, endothelial dysfunction, autonomic imbalance) anchored to the 2021 WHO 5 µg/m³ annual guideline. Adds little new; useful only as an introductory citation.

doi: 10.32394/pe/218966

Luo et al. 2026 Front Public Health

Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning

87,913 CVD deaths, 2014–2024; generalized additive quasi-Poisson adjusted for long-term trend, temperature, RH and day of week, across single-day lags 0–7 and cumulative windows 0–1 to 0–7. At lag 0, **RR 1.014 (1.008–1.020) for PM_{2.5}**, 1.009 (1.006–1.013) for PM₁₀ and 1.018 (1.011–1.026) for NO₂; SO₂ peaks at lag 1 with **1.044 (1.018–1.069)**, the largest coefficient in the set. Subtropical southwestern China is genuinely under-represented in this literature. Two-pollutant models were run, but the paper does not resolve the SO₂/PM_{2.5} collinearity its own results imply, so the apparent SO₂ dominance should not be read as source-specific.

doi: 10.3389/fpubh.2026.1878398

Rusconi et al. 2026 GeoHealth

Long-term ultrafine particle exposure 2010–2019 and cardiovascular burden in high-income countries

Applies a new global UFP dataset to compute population-weighted exposure for high-income countries and correlates it with country-level CVD DALYs, YLLs and YLDs, reporting “consistent patterns” between long-term UFP exposure and CVD burden. This is a country-level ecological correlation with no individual data, no lag structure and no control for the PM_{2.5} with which UFP is strongly collinear — the authors say so, restrict to high-income countries deliberately, and warn against generalisation. It cannot support causal or even population-attributable inference. Included only because it is the corpus's sole ultrafine-focused record, which is itself the point: UFP epidemiology still lacks the exposure infrastructure PM_{2.5} has.

doi: 10.1029/2025GH001636

Xi et al. 2026 Diabetologia

Air pollution, built environment and temperature vs. prediabetes phenotypes (KORA)

KORA cohort, Southern Germany 2006–2022, 1,618 participants, complementary log-log models plus quantile g-computation for mixtures. Per-IQR PM_{2.5} gave **HR 1.89 (1.07–3.31) for combined IFG+IGT**; lower NDVI HR 1.59 (1.03–2.46). Crucially, associations were with *incidence and non-remission* of impaired glucose tolerance rather than progression to type 2 diabetes — a phenotype-resolution result that most air-pollution/diabetes studies cannot deliver.

doi: 10.1007/s00125-026-06814-2

Yang et al. 2026 J Am Heart Assoc

Short-term PM_{2.5} and hospitalisation/ED use in atrial fibrillation

2021 Medicare data, 26,158 western-US patients with established AF; ZIP-code daily PM_{2.5} from a 1 km geospatial model, threshold ≥9 µg/m³, lags 0/1/3 d. **OR 1.10 (1.01–1.20) for CVD hospitalisation; 1.24 (1.03–1.48) for AF hospitalisation; 1.34 (1.06–1.70) for AF ED visits**. No significant PM×extreme-heat interaction — a negative finding worth noting given how often synergy is assumed.

doi: 10.1161/JAHA.125.046917

Zhao et al. 2026 J Am Coll Cardiol

Compound extreme heat and fire-sourced air pollution vs. cardiovascular mortality

Multi-country time-series across six countries, 5.9 million CVD deaths, with each day classified into mutually exclusive states (no hazard / heat-wave only / substantial fire-sourced air pollution only / compound) and quasi-Poisson RRs plus formal additive interaction. **Compound RR 1.115 (1.100–1.130)** vs. heat-only 1.083 (1.074–1.091) and SFAP-only 1.014 (1.008–1.021); RERI 0.018 (0.002–0.035), attributable proportion 1.598% (0.181–3.129). The interaction is statistically present but small — most compound risk is the heat term — and the mutually-exclusive day classification discards dose information within each state. Fire-sourced attribution rests on modelled smoke fields, so exposure misclassification is non-differential but real. Sets the design standard for compound-hazard work.

doi: 10.1016/j.jacc.2026.06.005

Zhou et al. 2026 iScience

Air pollution and hypertension: an umbrella review of systematic reviews and meta-analyses

Umbrella review over three databases to September 2025, AMSTAR-2-appraised, retaining eleven meta-analyses (eight on essential hypertension, three on gestational hypertension). Conclusion: cumulative evidence supports PM_{2.5} and PM₁₀ raising essential hypertension risk, while the gestational-hypertension association remains inconclusive on three studies. The value of this record is as a citation anchor and a map of between-review heterogeneity; the abstract reports **no pooled estimate, no I², and no AMSTAR-2 distribution**, so the strength of the “plausible role” claim cannot be checked without the full text. Umbrella reviews also inherit the primary literature's exposure-model heterogeneity —

land-use regression, satellite-derived and nearest-monitor assignments are pooled as though interchangeable, which is precisely the assumption a co-located sensor network is positioned to test.

doi: [10.1016/j.j.isci.2026.116811](https://doi.org/10.1016/j.j.isci.2026.116811)

Neuro / mental health

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 13 RECORDS IN THIS SUBTOPIC)

Calderón-Garcidueñas et al. 2026 J Alzheimers Dis

Abnormal eye movements reflect early cortical and volumetric brain changes in PM_{2.5}-exposed urban youth

Video-based eye tracking (EyeLink 1000-plus) in 80 volunteers aged 33±11 from Metropolitan Mexico City and lower-exposure Cuernavaca, with structural MRI in 45 MMC subjects. **Oculomotor variables did not differ significantly between cities** — the headline null. Abnormal antisaccades, low pursuit gain and square-wave jerks were merely *more frequent* in MMC, and the significant findings are correlations between oculomotor measures and frontal-temporal-parietal cortical, hippocampal, thalamic, caudate, amygdalar and cerebellar structure *within* the exposed group. With $n = 45$ and that many regions the multiple-comparison exposure is severe, and no correction is described. The group's long-running MMC series is valuable, but this instalment does not establish an exposure contrast.

doi: [10.1177/13872877261471048](https://doi.org/10.1177/13872877261471048)

Cattarinussi et al. 2026 medRxiv

Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories

ABCD Study, $n = 10,620$, ages 9–16, sex-stratified trajectories of internalising symptoms and psychotic-like experiences against residential PM_{2.5} and polygenic scores for schizophrenia and major depression. PM_{2.5} was associated with **smaller decreases in PLEs over time in females** ($p\text{-FDR} = 0.005$), with no effect on internalising trajectories in either sex, and a **PM_{2.5} × PRS-MDD interaction on PLE trajectories in females** ($p\text{-FDR} = 0.048$) — a borderline interaction that will not survive replication unless the effect is real. PM_{2.5} is residential-address-assigned, so within-person exposure misclassification is large and non-differential, which biases toward the null for main effects but has unpredictable sign for interactions. Preprint.

doi: [10.64898/2026.07.12.26357864](https://doi.org/10.64898/2026.07.12.26357864)

Dales et al. 2026 JAMA Netw Open

Wildfire-sourced PM_{2.5} and ED visits for migraine and other primary headache

Case-crossover over 997,701 ED visits, 2010–2023 Alberta/Ontario, using the Canadian Optimized Statistical Smoke Exposure Model to separate wildfire-sourced from background PM_{2.5}. **+6.07% (5.75–6.39) per wildfire exposure**; background PM_{2.5} null. Effect modification by material/social deprivation is large (3.61% → 10.42% across quintiles). The methodological contribution — explicit source attribution within a case-crossover frame — is more transferable than the headache endpoint itself.

doi: [10.1001/jamanetworkopen.2026.25015](https://doi.org/10.1001/jamanetworkopen.2026.25015)

Daoud et al. 2026 Environ Sci Pollut Res Int

Air pollution and suicide mortality in the UK

Balanced panel of **>2,200 local-authority-year observations, 2015–2023**, linking ONS suicide data to DEFRA modelled PM_{2.5} with local-authority and year fixed effects, plus nonlinear specifications. The raw cross-sectional association is *negative*; after fixed effects it reverses to $\beta = 0.221$ ($p = 0.007$), with a U-shaped form turning at $\approx 7.8 \mu\text{g}/\text{m}^3$. That sign reversal is the whole result and it is not reassuring: the estimate is driven entirely by within-area variation over a period of falling concentrations and rising suicide rates, with no time-varying socioeconomic controls (deprivation, unemployment, service provision) in the model. Ecological, modelled exposure, no individual-level confounders. The authors describe it as exploratory, which is the correct reading. Logged here because a second, independent PM–suicide record surfaced in the same window (Cai et al., below).

doi: [10.1007/s11356-026-38089-w](https://doi.org/10.1007/s11356-026-38089-w)

Glover et al. 2026 Age Ageing

Umbrella meta-review of urban environmental exposures and cognitive health

54 systematic reviews, 66 meta-analysed outcomes, 270 exposure–outcome associations, graded by sample size, effect strength and bias. PM_{2.5} is the strongest-graded exposure (10/11 significant outcomes, “highly probable” detrimental), ahead of NO₂/NO_x (5/10). Social relationships were probably protective; green/blue space associated with lower dementia risk (3/6). Useful as a citation anchor for exposure hierarchy claims.

doi: [10.1093/ageing/afag215](https://doi.org/10.1093/ageing/afag215)

Goossens et al. 2026 Neurosci Appl

In utero air pollution exposure and pre-/postnatal brain development

Imaging-focused review (fetal ultrasound + childhood/adolescent MRI) stratified by pollutant. Prenatal ultrasound studies in high-exposure regions show reduced biparietal diameter and head circumference; childhood MRI in low-exposure regions shows no global volume effect but regional cortical, subcortical grey and white matter differences. The exposure-contrast dependence of the discrepancy is the point worth carrying forward.

doi: [10.1016/j.j.nsa.2026.107019](https://doi.org/10.1016/j.j.nsa.2026.107019)

Kabir et al. 2026 Health Place

Environmental attributes as moderators of neighbourhood deprivation and mental health

Systematic review across seven databases plus grey literature, 19 studies included, asking specifically whether objectively measured environmental attributes moderate the deprivation–mental-health association. Moderation findings were mixed: greenness and blue space generally protective in disadvantaged neighbourhoods, air pollution exacerbated poor mental health, parks and walkability inconsistent. Noise and elder-support centres were each examined in exactly one study. With 19 heterogeneous studies and no meta-analysis, the moderation claim is directional at best, and the authors' call for longitudinal designs is the honest conclusion. Relevant if you intend to use LCS network output as an effect modifier rather than a main exposure.

doi: [10.1016/j.healthplace.2026.103719](https://doi.org/10.1016/j.healthplace.2026.103719)

Lu et al. 2026b Environ Res

Subchronic PM_{2.5} exposure exacerbates chronic stress-induced depression-like pathology in mice

Factorial murine design crossing subchronic PM_{2.5} inhalation with chronic unpredictable mild stress (CUMS), read out by behavioural batteries, hippocampal histopathology, transcriptomics, electrophysiology and synaptic-plasticity markers. The result that matters is the **interaction, not the main effect**: PM_{2.5} alone produced no depression- or anxiety-like behaviour, only modest neurobiological change, whereas PM_{2.5} + CUMS

worsened anhedonia, behavioural despair and anxiety relative to CUMS alone, with greater hippocampal neuroinflammation, redox disruption and impaired synaptic plasticity. This supports an effect-modifier rather than a direct-cause model. The abstract reports no exposure concentration, no particle characterisation and no effect sizes, so dose–response is unassessable and the inhalation regime cannot be compared with ambient exposure; “real-ambient” status is not claimed. For sensing, the implication is that a monitoring design aimed at mental-health endpoints needs to resolve *co-occurrence* of exposure and psychosocial stressor, which is a personal-monitoring problem, not a fixed-site one.

doi: 10.1016/j.envres.2026.125327

Ma & Yau 2026 iScience

PM_{2.5}-induced neurotoxicity and depression: exposure risks and mechanisms

Narrative integration of epidemiological, animal and mechanistic evidence. Component-focused: metals, black carbon and organic compounds are implicated via oxidative stress, neuroinflammation, blood–brain barrier disruption, HPA-axis dysregulation and kynurenine-pathway alteration. Explicitly calls for component-resolved exposure assessment — relevant if your instrumentation work can deliver speciated output.

doi: 10.1016/j.isci.2026.116617

Pan et al. 2026 Zhonghua Yu Fang Yi Xue Za Zhi

Maternal PM exposure and offspring neurodevelopment: effects and mechanisms

Chinese-language review covering placental barrier disruption and particle translocation, oxidative stress, epigenetic alteration, endocrine dysfunction and gut microbiota imbalance. Largely confirmatory of the above; included for completeness of the mechanistic map.

doi: 10.3760/cma.j.cn112150-20250804-00752

Sarkar et al. 2026 J Appl Toxicol

PM-induced neurotoxicity: neuroinflammation and organelle dysfunction

Narrative review mapping CNS entry routes (olfactory-bulb translocation, blood–brain barrier disruption, peripheral inflammatory signalling) onto subcellular failure modes: unfolded protein response with CHOP and caspase-12 induction, impaired oxidative phosphorylation, ROS accumulation, mitochondrial permeability transition, and compromised autophagic flux and lysosomal integrity. Coherent as a mechanistic map but adds no new data, and — as with most of this literature — draws heavily on instillation and high-concentration *in vitro* models whose dosimetry does not correspond to ambient exposure. Use for the organelle-level framing, not for dose–response.

doi: 10.1002/jat.70360

Zhang et al. 2026 Neurotoxicology

PM_{2.5} and its components and cognitive dysfunction among rural older adults

14,379 Henan Rural Cohort participants aged ≥60, cognitive dysfunction defined by MMSE and the Hasegawa Dementia Scale, with PM_{2.5}, black carbon, organic matter, ammonium, sulfate and nitrate from the TAP dataset; logistic and weighted quantile sum regression, then network toxicology and random forest for candidate genes. **Per-IQR increases raised CD odds by 19–43% across the six exposures**, and **BC was the leading mixture contributor under all four CD definitions** (weights 54–86%), with OM second (7–35%). Cross-sectional, so reverse causation through cognitively impaired participants’ residential and behavioural patterns is unaddressed, and component fields from a single reanalysis product are strongly collinear — WQS weights under collinearity are unstable. The BC ranking is consistent with the emerging component literature; the network-toxicology arm is hypothesis-generating and labelled as such by the authors.

doi: 10.1016/j.neuro.2026.103539

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Cai et al. 2026 Atmos Environ

PM_{2.5} as the dominant air pollutant associated with suicide among people living with HIV/AIDS: province-wide case-crossover

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122271

Respiratory & allergic

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 13 RECORDS IN THIS SUBTOPIC)

Bayar Muluk 2026 Int J Biometeorol

Climate change and otolaryngology: airway disease and surgical emission reduction

Dual-domain narrative review. The relevant claim is a proposed *synergy* between elevated CO₂ and PM_{2.5} increasing pollen allergenicity and mucosal penetration. The second half (operating-room carbon footprint) is outside this watch’s scope.

doi: 10.1007/s00484-026-03289-z

Chaiwong et al. 2026 Int J Mol Sci

PM_{2.5} and systemic inflammatory biomarkers in elderly COPD versus healthy elderly

Pilot panel of 38 COPD patients (73.6 ± 7.1 y) and 20 age-matched controls, sampled in a high- (79.1 ± 23.3 µg/m³) and a low-pollution (13.9 ± 3.6 µg/m³) Chiang Mai season, measuring hsCRP, IL-6, IL-8 and TNF-α. IL-8 was higher in COPD across both periods, but **no significant within-group increase accompanied the 5.7-fold seasonal exposure contrast** — a null the authors attribute to inhaled corticosteroids (65.8% of COPD) and statins (42% COPD, 65% controls). That explanation is untestable at *n* = 58; equally consistent are ceiling effects on chronic inflammation, ambient-monitor exposure misclassification, and simple lack of power. Reported as a pilot, and should be cited as one. Relevance: a seasonal contrast this large failing to move systemic biomarkers argues for personal rather than municipal exposure metrics in susceptible-population panels.

doi: 10.3390/ijms27146283

AIRCLARE-Ghana 2026 (preprint) Research Square

Short-term ambient PM_{2.5}/PM₁₀ and acute respiratory morbidity in Ghana

See *Signal of the day*. Time-stratified case-crossover, Jan 2023–Dec 2024, 9,511 acute respiratory presentations from 213,241 outpatient records, conditional logistic regression adjusted for temperature and relative humidity. **URI: OR 1.038 (1.010–1.068) per IQR PM_{2.5}; OR 1.030 (1.008–1.052) per IQR PM₁₀**, peaking at lag 1 and persisting to cumulative lag 0–7 d. Pneumonia null, asthma null. Threat to inference: one monitoring site stands in for a whole catchment, so Berkson-type and classical exposure error coexist and the true effect is plausibly larger; the case definition is bespoke and unvalidated; and it is a preprint. The 96.03% post-QC completeness figure is the number a network operator should take away.

doi: 10.21203/rs.3.rs-10441248/v1

Li et al. 2026 mSystems

Host genetic variation, airway microbiome and environmental exposure vs. respiratory health

Sputum metagenomes from 1,651 individuals, with Mendelian randomisation implicating SO₂, CO and PM_{2.5} effects on lung function potentially mediated by *Actinomyces*, *Haemophilus influenzae* and *Veillonella* spp., and genotype-dependent exposure–microbiome interactions (*MEOX1* with lower *Filifactor alocis* under pollutant exposure; *FAM110D* and *USP36* with higher *Ralstonia pickettii*/*Neisseria* and lower *Oribacterium*, tracking COPD risk). Fungal loci also reported. This is many hypotheses from one cohort: MR on aggregate exposures with microbial mediators is fragile to pleiotropy and the variant identifiers are rare or novel, so the loci need replication before they mean anything. Mechanistically suggestive for why identical exposure yields divergent lung-function trajectories.

doi: [10.1128/msystems.00442-26](https://doi.org/10.1128/msystems.00442-26)

Liao et al. 2026 Med Gas Res

Short-term air pollution, respiratory symptoms and lung function in asthma

Small panel study (32 patients, 266 visits, Beijing 2015–2016) with mixed-effects models. Per-IQR PM_{2.5} (74.5 µg/m³, 6-day) reduced Asthma Control Test scores 3.6%; PM₁₀ 7-day moving average reduced FEV₁% predicted by 3.6% and FEF₂₅% by 6.0%. Exposure levels are far above current WHO guidance — limits extrapolation to low-exposure settings.

doi: [10.4103/mgr.MEDGASRES-D-25-00164](https://doi.org/10.4103/mgr.MEDGASRES-D-25-00164)

Mokoena et al. 2026 Public Health Chall

Socio-economic conditions, air pollution sources and eczema symptoms in Soshanguve teenagers

Cross-sectional modified ISAAC III survey of 2,885 schoolchildren aged 13–14 across 13 schools in Tshwane, South Africa; multiple logistic regression. Current eczema symptoms 26.8%, eczema ever 23.3%. Frequent truck traffic near the home: AOR 1.26 (current) and 1.24 (ever); ETS 16–20 days/month: AOR 1.96; female sex, monthly paracetamol use (AOR 2.37) and pet ownership also associated. No confidence intervals are reported for most estimates, exposure is entirely self-reported with no measurement, and the paracetamol association is the textbook reverse-causation trap. Value is geographic coverage — one of very few sub-Saharan records in this corpus — not effect estimation.

doi: [10.1002/puh2.70325](https://doi.org/10.1002/puh2.70325)

Sánchez de Prada et al. 2026 Epidemiologia

Ambient PM and asthma-coded admissions in children aged 0–3, Spain

Nationwide municipality-year ecological panel, 2020–2022, 20,746 municipality-years, 3,231 admissions — but 93.7% of municipality-years are zero, which is the dominant statistical feature. Poisson IRRs per IQR: PM_{2.5} **1.54 (1.28–1.85)**, PM₁₀ **1.89 (1.59–2.26)**; attenuated but persistent under negative-binomial and province fixed effects. Multipollutant models were collinear and unstable. Ecological design — do not read as individual-level risk.

doi: [10.3390/epidemiologia7040097](https://doi.org/10.3390/epidemiologia7040097)

Wang et al. 2026 Front Med

Environmental and socioeconomic influences on childhood asthma and pneumonia burden

Retrospective cohort from paediatric electronic medical records at one Wuhu hospital, Anhui, January 2022–December 2025: 936 children aged 1–14 (47.4% asthma, 36.5% pneumonia, 16.0% both), with regional NO₂, PM_{2.5}, temperature and humidity and census-based socioeconomic status. Higher NO₂ and PM_{2.5} were associated with asthma; biomass fuel use and poor household ventilation with pneumonia; lower SES strongly with pneumonia but not asthma. Pneumonia peaked in winter, asthma in spring. Hospital-based case series with area-level exposure and no comparison population — the associations describe who among the diagnosed had which condition, not incidence.

doi: [10.3389/fmed.2026.1872723](https://doi.org/10.3389/fmed.2026.1872723)

Yang et al. 2026 J Hazard Mater

Allergenic vs. toxic risk of airborne seed hairs compared with ambient PM_{2.5}

Simultaneous urban/suburban sampling of poplar-willow seed hairs and PM_{2.5} in Hohhot, May 2025, with sIgE-based allergenic quantification and FVOC/heavy-metal toxicity. Seed hairs carry 13.1±12.1× the pollen and 92.9±55.2× the bacterial load per unit mass, giving sIgE levels 14.9× and 124× higher than concurrent PM_{2.5}; conversely PM_{2.5} FVOC toxicity exceeds seed hairs by ~4×10⁴. A clean demonstration that mass-based monitoring misallocates seasonal allergenic risk entirely.

doi: [10.1016/j.jhazmat.2026.143076](https://doi.org/10.1016/j.jhazmat.2026.143076)

Yue et al. 2026 Front Public Health

Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou

20,623 paediatric bronchopneumonia admissions (ICD-10 J18, ages 0–14) at a single tertiary hospital in Huadu District, Guangzhou, 2014–2023, against daily NO₂ and SO₂ in a quasi-Poisson generalised additive model. Per 5 µg/m³, peak single-day excess risk at lag 4 was **2.74% (95% CI 1.73–3.75)** for NO₂ and **8.81% (4.94–12.82)** for SO₂; over the 0–7-day moving average these rose to **6.83% (4.93–8.77)** and **17.21% (10.18–24.70)**. Children aged 6–14 and boys were consistently more susceptible. Two problems: no particulate term is reported at all, so the gaseous estimates absorb whatever correlated PM signal exists and cannot be read as pollutant-specific; and a single-hospital catchment in a ten-million-person megacity leaves the denominator undefined and exposed to referral-pattern drift over a decade. Included because the lag structure and the age–sex susceptibility gradient are informative, not because the point estimates are attributable.

doi: [10.3389/fpubh.2026.1885590](https://doi.org/10.3389/fpubh.2026.1885590)

Zhang et al. 2026 Epidemiology

Infancy air pollution exposure and atopic dermatitis at 4–6 years

Pooled longitudinal analysis of 1,587 children across six US cities in the ECHO-PATHWAYS Consortium (2017–2023); first-year-of-life PM_{2.5}, NO₂ and O₃ from validated national spatiotemporal models; three caregiver-reported outcomes; Poisson models with prespecified modification by child sex and maternal education. **PM_{2.5} and NO₂ null**; O₃ associated with current AD (aRR 1.07 per 1 ppb, 1.01–1.13) and location-specific AD (1.06, 0.99–1.13), evident only in children of lower-educated mothers. Outcome ascertainment is caregiver questionnaire, not clinical examination, and the education-stratified finding is a subgroup result on a modest sample. A well-powered PM_{2.5} null against a validated exposure model is worth as much as a positive.

doi: [10.1097/EDE.0000000000002033](https://doi.org/10.1097/EDE.0000000000002033)

Zhu et al. 2026 Medicine (Baltimore)

Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis

Five years of pneumonia admissions (2015–2019) in an inland Chinese energy city against six pollutants, generalised additive plus distributed-lag nonlinear models. Per 10 µg/m³: **PM_{2.5} RR 1.004 (0.991–1.017)** and **PM₁₀ 1.004 (0.999–1.010)** — both null — against NO₂ **1.221 (1.174–1.271)**, **SO₂ 1.035 (1.022–1.048)**, and CO 1.382 (1.230–1.552) per 1 mg/m³. Effects peaked at lag 1 for PM_{2.5} and lag 0–7 for the gases; men were more vulnerable to SO₂, women to CO, and children ≤14 formed the high-risk group. The single-pollutant framing is the limitation — in a heating-dominated city PM_{2.5}, SO₂, NO₂ and CO share a common combustion source and a common seasonal cycle, and no multi-pollutant

model is presented, so the apparent NO₂ dominance is a statement about correlation structure rather than about which pollutant acts. Useful mainly as a demonstration that PM mass can be the null term in its own time-series.

doi: 10.1097/MD.00000000000049962

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Lu et al. 2026 Atmos Environ

Antagonistic effects of PM_{2.5} and ozone on respiratory morbidity: multi-pollutant interaction, Pearl River Delta
No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122270

Reproductive & developmental

10 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 10 RECORDS IN THIS SUBTOPIC)

Bello et al. 2026 Econ Hum Biol

Prenatal PM_{2.5} exposure and infant health in Quebec

Administrative birth records for Quebec 2008–2015 linked to monitoring data and tax records, in neighbourhood-by-quarter fixed-effects models — a low-pollution setting, which is the point. **Population-average effects on birth outcomes were essentially null**; the result lives entirely in the heterogeneity. Among families in the most vulnerable stratum of a cumulative socioeconomic vulnerability index, a **1 µg/m³ increase in mean PM_{2.5} was associated with ≈6.6 g lower birth weight**, with further effects on prematurity and birth weight among newborns admitted to neonatal care. **Days exceeding the Canadian 24 h standard had disproportionately large effects in vulnerable families**, so peaks matter beyond the mean — directly relevant to any monitoring design optimised for annual averages. No confidence intervals are reported in the abstract, and the vulnerability index is constructed rather than pre-registered, which invites specification search across strata.

doi: 10.1016/j.ehb.2026.101631

Chen et al. 2026 J Environ Sci (China)

Prenatal air pollution, fetal growth trajectories and child neurodevelopment

Time-series *k*-means clustering to derive fetal growth trajectory phenotypes, then mediation analysis. Sensitive windows identified at 0–20 and 27–41 weeks; higher exposure raised odds of high-risk trajectories by 9–32%, and first-trimester crown-rump length plus mid-pregnancy estimated fetal weight mediated effects on the 2-year psychomotor development index. The trajectory-clustering approach is the methodologically interesting part.

doi: 10.1016/j.jes.2025.10.043

Huang et al. 2026 Ecotoxicol Environ Saf

Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ

Guangxi Zhuang Birth Cohort, 231 mother–child pairs, WISC indices, analysed by generalized linear models, restricted cubic splines, PCA, quantile g-computation and BKMR. Prenatal PM_{2.5} and PM₁₀ associated with lower verbal comprehension and full-scale IQ; effects stronger in boys; BKMR shows an inverted-U for fluid reasoning against the 10-pollutant mixture. Small *n* — treat the mixture results as hypothesis-generating.

doi: 10.1016/j.ecoenv.2026.120566

Li et al. 2026 Ecotoxicol Environ Saf

Ambient air pollutant exposure and semen quality: a refined multi-window panel study

One-year prospective panel of 32 male undergraduates aged 18–22 with four quarterly follow-ups spanning the 90-day spermatogenic cycle, using both daily-average and cumulative-dose exposure metrics and — the methodological contribution — stratifying the 15–69-day meiotic window into three biologically distinct sub-stages. Linear mixed-effects, two-pollutant and Bayesian weighted quantile sum models. Per IQR of full-cycle O₃, **sperm concentration fell 15.45% (95% CI –20.95 to –9.58)**, with the sub-stage peak at days 31–45 (–13.30%, –18.09 to –8.24); NO₂ acted earliest (15–30 d), CO later (31–69 d). With *n* = 32 the sub-stage resolution outruns the power, and homogeneous undergraduates limit generalisability by design. The window-stratification framework is worth borrowing regardless.

doi: 10.1016/j.ecoenv.2026.120581

Lin et al. 2026 Environ Pollut

Maternal PM exposure during pregnancy and offspring liver function

Fujian Birth Cohort, 1,046 mother–offspring pairs, with gestational and trimester-specific PM₁, PM_{2.5} and PM₁₀, serum ALT and AST at ages 1–2 years, restricted cubic splines for shape and distributed lag models for timing. **Second-trimester IQR increments gave ALT +1.42 (0.82–2.03), +1.41 (0.77–2.05) and +0.91 (0.35–1.46) U/L** for PM₁/PM_{2.5}/PM₁₀, with AST increases of similar magnitude. The effect sizes are small relative to the assay's biological variability in toddlers and clinical significance is unestablished; a single post-natal measurement precludes trajectory inference, and PM₁/PM_{2.5}/PM₁₀ are too collinear to separate as cleanly as the paper implies. The distributed-lag identification of a second-trimester window is the durable part.

doi: 10.1016/j.envpol.2026.128835

Liu et al. 2026 Int J Mol Sci

Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models

Head-to-head sensitivity comparison of HTR-8/SVneo and BeWo trophoblast lines against three maternal-exposome stimuli — the mycotoxin deoxynivalenol, a TNF-α/IL-1β/IFN-γ cocktail, and PM_{2.5} — at two penicillin/streptomycin concentrations, 24 h exposure. **The PM arm was null**: no change in IL-6, IL-8 or hCG in either line, while DON and the cytokine cocktail moved IL-6/IL-8, hCG and junctional markers (ZO-1, OCLD, CLDN-3/4, E-CAD). A negative result in a 24 h submerged monoculture at an unreported dose is weak evidence of placental safety and strong evidence that these two lines are the wrong model for particle toxicity — there is no air–liquid interface, no deposition physics and no particle characterisation. Worth logging precisely because prenatal-PM epidemiology keeps citing trophoblast inflammation as its mechanism.

doi: 10.3390/ijms27146434

Pai et al. 2026 Pediatr Neonatol

Household incense burning and small-for-gestational-age risk

18,666 mother–infant pairs in the Taiwan Birth Cohort Study, incense exposure during pregnancy categorised none / occasional / daily, SGA stratified by term and preterm. SGA prevalence rose monotonically across exposure groups (8.1%, 9.4%, 10.0%). Among term infants, **aOR 1.16 (1.00–1.33) occasional and 1.25 (1.10–1.41) daily**; preterm infants showed the same direction without significance, and there was no association with preterm birth itself. Sensitivity analyses covered other indoor environmental factors, maternal medical history, gestational complications,

gestational weight gain and secondhand-smoke interaction. The limitation that matters is that exposure is a self-reported frequency category with no concentration, duration or ventilation information — the dose–response is ordinal in *practice*, not in dose, and the contrast between an unventilated daily burn and a well-ventilated one sits entirely inside the top category. Pairs directly with Zhou L et al. in this issue.

doi: 10.1016/j.pedneo.2026.06.003

Peterson 2026 Curr Environ Health Rep

Ambient air pollution, PAHs and pubertal development: critical review

Component-level synthesis. Organic matter and sulfate fractions of PM_{2.5} are associated with earlier menarche and precocious puberty; OM is enriched in combustion-derived PAHs. Notably, high *chronic* exposure in heavily polluted settings has also been associated with *delayed* menarche — a non-monotonic pattern that any dose–response modelling should accommodate. Proposed mechanism: disruption of hypothalamic kisspeptin–GnRH signalling.

doi: 10.1007/s40572-026-00552-8

Sezer et al. 2026 Birth Defects Res

Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate

Hospital-based case-control study, 127 nonsyndromic CLP cases against 271 controls, with PM_{2.5}, PM₁₀, O₃, NO_x, SO₂ and CO assigned for the three months before conception and the first trimester. Higher O₃, PM_{2.5} and PM₁₀ were each associated with elevated CLP risk in single-pollutant models; **O₃ was the dominant contributor in the WQS mixture** and the only exposure stable across windows and model specifications, while traffic-related pollutants did not survive multi-pollutant adjustment. The study is underpowered for a mixture analysis — 127 cases against a six-pollutant WQS index invites weight instability, and no bootstrap weight distribution is reported — and hospital-based control selection risks referral bias. Read as hypothesis-generating. The methodological point that carries: with correlated pollutants and a small case series, the pollutant that wins the mixture is often the one with the widest exposure contrast, not the one with the strongest biology.

doi: 10.1002/bdr2.70102

Xie et al. 2026 Environ Res

Long-term PM_{2.5} and timing of pubertal development in boys and girls

Chongqing Child Health Cohort 2014–2024, 3,833 children, time-varying cumulative average residential PM_{2.5} from the prenatal period, Cox models across nine milestones. **Boys: testicular HR 1.065 (1.056–1.073), penile 1.063 (1.055–1.071). Girls: breast 1.044 (1.039–1.050), menarche 1.033 (1.025–1.040)**, all per 1 µg/m³. Associations were stronger in overweight/obese girls and in areas with higher long-term temperature and lower humidity — i.e. meteorological effect modification, not just confounding. The male-inclusive design is the field's main gap being closed here.

doi: 10.1016/j.envres.2026.125313

Mechanistic toxicology

11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 11 RECORDS IN THIS SUBTOPIC)

Balu et al. 2026 J Biochem Mol Toxicol

SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction

Langendorff-perfused male Wistar hearts exposed to NIST SRM 1650b heavy-duty diesel particulate at 10, 100 and 300 µg/mL before 30 min ischemia and 60 min reperfusion. Dose-dependent deterioration of haemodynamics and elevated tissue injury, with increased oxidative stress in tissue and isolated mitochondria, reduced bioenergetic enzyme activity and respiratory efficiency, downregulated *Pgc-1α*, *Tfam*, *Polg*, *Fis1*, *Mfn1* and *Pink1*, lower mtDNA copy number, and suppressed PI3K/Akt signalling that fell further after the ischemic challenge. The design isolates a direct cardiomyocyte effect cleanly — no autonomic, pulmonary or systemic inflammatory route is available in an ex-vivo preparation — which is both its strength and its ceiling. The concentrations are far above any inhaled dose delivered to myocardium in vivo, and direct coronary perfusion bypasses the alveolar barrier entirely, so this establishes a mechanism that *can* operate, not one that does at ambient exposure.

doi: 10.1002/jbvt.71038

Ding et al. 2026 J Environ Sci (China)

PM_{2.5} promotes vascular calcification via miR-27b-5p/HES1 signalling

Mouse aortic smooth-muscle cells, hiPSC-derived 3D blood-vessel organoids, and a murine model. Proposed axis: PM_{2.5} suppresses miR-27b-5p transcription in an XBP1-dependent manner → HES1–Runx2 interaction → osteogenic transformation of VSMCs. miR-27b-5p mimics rescued the organoid phenotype. The organoid arm is the credible translational bridge; dose and duration dependence shown in vivo.

doi: 10.1016/j.jes.2025.12.074

Fakfumi et al. 2026 Toxics

Seasonal PM_{2.5} exposure and plasma metabolome changes related to metabolic syndrome

Prospective within-subject comparison of plasma ¹H-NMR metabolomes in 25 healthy adults in Samoeng District, Chiang Mai, between haze and non-haze seasons. Twenty-six metabolites separated the seasons by PLS-DA (VIP > 1.5): eleven up (maleylacetoacetic acid, deoxyribose 5-phosphate, betaine, 3-hydroxyanthranilic acid, 1-methyladenosine) and fifteen down (deoxyguanosine, D-arabitol, glycerophosphocholine, ophthalmic acid, oxaloacetic acid), implicating inflammation, ROS generation, NO inhibition and disrupted energy metabolism across 56 pathways. The design cannot separate PM_{2.5} from everything else that changes with the Thai dry season — temperature, humidity, diet, activity, and co-emitted gases — and PLS-DA at *n* = 25 over hundreds of features is prone to overfitting without the permutation testing the abstract does not report. Hypothesis-generating.

doi: 10.3390/toxics14070544

Liu et al. 2026 Environ Toxicol Pharmacol

Resveratrol ameliorates PM_{2.5}-aggravated IBS-D via microbiota and metabolite restoration

Three-arm murine design (IBS-D, PM_{2.5}+IBS-D, resveratrol+PM_{2.5}+IBS-D) with untargeted metabolomics, 16S microbiota profiling, colonic histopathology, oxidative-stress and barrier-function assays and visceral-hypersensitivity testing. PM_{2.5} worsened IBS-D with dysbiosis and disrupted hypoxanthine and linoleic-acid metabolism; resveratrol restored both. No exposure concentrations, particle source or dose–response are reported in the abstract, group sizes are unstated, and the resveratrol arm is a single dose — so this is proof-of-concept, not a therapeutic result. The gut-axis endpoint remains underexplored, which is why it is retained.

doi: 10.1016/j.etap.2026.105120

Lu et al. 2026 Toxics

Toxicity of PM_{2.5} from incomplete solid-fuel burning in Caenorhabditis elegans

Synchronised L4 *C. elegans* exposed to aqueous suspensions of PM_{2.5} collected from incomplete combustion of six solid fuels (rice, wheat, peanut and rapeseed straw; poplar and paulownia branch), scored for body length, fertilised-egg count, lipofuscin accumulation and ROS, alongside OC/EC, water-soluble ion and PAH characterisation. **Oilseed-crop straw (peanut, rapeseed) was the most toxic**, and the effect tracked lipid-soluble PAH content via oxidative stress. The source-specificity result is the useful part — it says fuel type, not mass concentration, drives toxicity, which is an argument for speciation-capable monitoring. The model is the limitation: an aqueous suspension delivered to a nematode cuticle and gut has no relationship to pulmonary deposition, and no mammalian or human-cell confirmation is offered.

doi: [10.3390/toxics14070597](https://doi.org/10.3390/toxics14070597)

Maraschi et al. 2026 J Comp Physiol B

Osmoregulatory disruption from settleable atmospheric PM in estuarine fish

96 h exposure of *Centropomus parallelus* to 1 g L⁻¹ settleable APM. Al, Fe, V and Cr accumulated in gill and kidney; branchial Na⁺/K⁺-ATPase activity and NKA⁺ ionocyte density rose while renal ATPase activity fell. Ecotoxicological rather than human-health relevant, but a useful reminder that deposition, not inhalation, is the dominant PM pathway in aquatic receptors.

doi: [10.1007/s00360-026-01698-5](https://doi.org/10.1007/s00360-026-01698-5)

Mishra et al. 2026 Atmos Chem Phys

Aerosol oxidative potential and reactive species predicted with a chemical kinetics model (KM-OP)

A chemical kinetic model of oxidative potential built on literature rate coefficients and a large compilation of laboratory data for transition-metal ions, quinones and organic aerosol, predicting ROS production and consumption of ascorbic acid and DTT from PM composition. Applied to field measurements from **Grenoble, Paris and London: $R^2 > 0.75$ with $< 15\%$ model bias in 4 of 6 datasets. Organic aerosol is the strongest contributor to antioxidant-based OP assays**, with minor Cu and Fe contributions; Cu dominates H₂O₂ production but barely affects •OH, while the antioxidant assays correlate well with modelled •OH — separating the species that drives the assay from the species that drives the metal signal. Two of six datasets are not reproduced, and the model is calibrated on the same body of literature it is tested against in kind. Even so, this converts OP from a per-filter measurement into a quantity predictable from speciated composition, which is exactly what a composition-resolving sensor network would need to be worth deploying.

doi: [10.5194/acp-26-10605-2026](https://doi.org/10.5194/acp-26-10605-2026)

Pan et al. 2026 Front Pharmacol

Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis

PRISMA systematic review and random-effects meta-analysis of 18 rodent studies, standardised mean differences for nasal mucosal markers. Air pollutant exposure was associated with increased eosinophils and elevated IL-4, IL-5, IL-13 and OVA-specific IgE — a coherent enhancement of type 2 inflammation — with scattered evidence of epithelial barrier loss (reduced ZO-1) and innate activation (IL-1 β , NLRP3). IFN- γ showed no consistent direction. Heterogeneity was substantial and the authors decline to claim a causal pathway. The pooling is the weak point: 18 studies spanning diesel exhaust particulate, PM_{2.5}, PM₁₀ and mixed urban aerosol at uncontrolled doses and durations, in an ovalbumin-sensitisation model whose relationship to human allergic rhinitis is itself contested, cannot yield a transferable effect size. No dose-response meta-regression is attempted, which is the analysis that would have justified the synthesis. Value here is as a marker map for the type 2 axis, not as an effect estimate.

doi: [10.3389/fphar.2026.1870023](https://doi.org/10.3389/fphar.2026.1870023)

Wang et al. 2026 Exp Eye Res

PM exposure, mitochondrial stress and ECM remodelling in scleral fibroblasts

Combined in vivo and in vitro: topically applied PM produced particle-surrogate distribution to ocular tissues with scleral thinning and axial elongation, while human scleral fibroblasts showed raised oxidative stress, mitochondrial ultrastructural abnormality, phenotypic change and altered collagen metabolism; transcriptomics implicated ECM organisation, mitochondrial metabolism and stress-response pathways. The framing as a myopia mechanism is plausible but the exposure route is the problem — topical application is not inhalation, and axial elongation in a rodent eye is a weak proxy for human myopia progression. Notable mainly as one of very few records addressing an ocular endpoint at all.

doi: [10.1016/j.exer.2026.111187](https://doi.org/10.1016/j.exer.2026.111187)

Wu et al. 2026 J Environ Sci (China)

Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation

Individually-ventilated-cage real-ambient exposure system, 15 weeks, plus transwell macrophage–hepatocyte co-culture and *Nlrp3* knockdown. Lysosomal damage and cathepsin B release → NLRP3 activation → IL-1 β → suppressed hepatocyte insulin signalling. Both NLRP3 inhibition and IL-1 β blockade rescued. Real-ambient exposure (rather than instillation) makes the dosimetry considerably more defensible than most of this literature.

doi: [10.1016/j.jes.2025.11.018](https://doi.org/10.1016/j.jes.2025.11.018)

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Saygin et al. 2026 Atmos Pollut Res

Source identification and seasonal change of trace metals, protein content and oxidative potential of fine and coarse particulates after the Kahramanmaraş earthquakes
No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: [10.1016/j.apr.2026.103164](https://doi.org/10.1016/j.apr.2026.103164)

Occupational & indoor

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 13 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Front Med Technol

Nested suction instrument to reduce intraoperative surgical-smoke PM_{2.5}

Bench study on porcine dorsal tissue with electrosurgical cutting, PM_{2.5} monitored by laser dust counters. Three orifice diameters tested (1.5/2.0/2.5 mm) with local smoke evacuation; reduction rates 24.2%, 66.9% and 43.0% respectively — the 2.0 mm “all-in-one” design (simultaneous smoke and blood suction) was optimal. Small, single-setting, but a clean engineering-control result.

doi: [10.3389/fmedt.2026.1796828](https://doi.org/10.3389/fmedt.2026.1796828)

Cole & Driscoll 2026 Ann Work Expo Health

Determinants of respirable crystalline silica exposure in Australian tunnelling

3,288 personal exposure measurements across 12 tunnel projects (2016–2024), obtained via freedom-of-information requests — the largest such dataset in Australian tunnelling. Linear mixed models explained 53–68% of variance. Underground workers had **2.3× the exposure of**

above-ground workers (GM 0.026 vs. 0.012 mg/m³); each 1% increase in silica fraction of respirable dust raised RCS exposure 5%; exposures fell after the July 2020 standard reduction (GM 0.023 → 0.013 mg/m³). A model for what transparent exposure-data access enables.

doi: 10.1093/annweh/wxag055

Ding et al. 2026 J Hazard Mater

Stage-resolved hair Zn-isotope dynamics as a firefighting exposure tracer

53 hair samples from 15 Beijing firefighters across five stage-defined periods. $\delta^{66}\text{Zn}$ ranged -0.79 to -0.13‰ , with non-linear incident–recovery trajectory: heavy-isotope enrichment after repeated missions, partial return toward baseline in late recovery. Linear mixed effects with stage-dependent Ni/Pb/Co associations; ICC 0.82 (i.e. dominated by between-subject variability). A genuinely novel time-integrated biomarker concept for combustion-derived PM.

doi: 10.1016/j.jhazmat.2026.143081

Ghosh et al. 2026 Indian J Occup Environ Med

Respiratory quality of life and pulmonary function in traffic police, eastern India

Cross-sectional comparison of 62 traffic police against 120 unexposed controls (Oct 2023–Jan 2024) using the St George's Respiratory Questionnaire and digital spirometry. Exposed personnel had significantly higher SGRQ scores and lower FVC and FEV₁ (both $p < 0.001$), with SGRQ correlating positively with pulmonary function measures. No exposure measurement whatsoever — occupation is the exposure proxy — and no adjustment for smoking, duration of service or socioeconomic position is reported, while simple random sampling from unequal groups invites selection differences. Reads as a hazard-surveillance signal for a large occupational group, not as an exposure–response result.

doi: 10.4103/ijoom.ijoom_150_25

Gwak et al. 2026 Risk Anal

Size-resolved modelling of airborne viral RNA copies in indoor environments

Size-resolved framework over 1–2000 μm coupling saliva-specific evaporation and residue formation with gravitational settling, across temperature, RH and two ambient pressures (101.325 and 75.3 kPa), combined with saliva-mass uncertainty, viral load, half-life, exhalation mode and ventilation. **Talking dominates airborne RNA persistence** because coughing and sneezing generate larger droplets that settle quickly. Predictions matched hospital air-sampling data within roughly one order of magnitude at the 50th saliva-mass percentile. The framework's own caveat is the binding one: RNA decay stands in for infectivity decay, and if viral RNA preferentially concentrates in long-suspended fine aerosol the size-resolved conclusions shift. Directly relevant to size-resolved optical sensing of bioaerosol in occupied spaces.

doi: 10.1111/risa.70318

Komonnithiphong et al. 2026 Toxics

PM_{2.5}-bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants

Ten area samples taken at customer dining tables in ten Chiang Mai charcoal-barbecue restaurants during peak evening hours in February 2026 (early dry season), 2 L min⁻¹ for 180 min onto quartz fibre filters, 16 US-EPA priority PAHs by GC-MS. Mean PM_{2.5} **87.64 $\mu\text{g m}^{-3}$** , total PAHs 132.74 ng m⁻³, carcinogenic fraction 52.3%, TEQ_{BaP} 15.34 ng m⁻³. Inhalation ILCR for a frequent adult diner **2.82×10^{-5}** (child 2.12×10^{-5}), rising to 6.16 and 4.62×10^{-5} at the worst site; even monthly diners exceed 10^{-6} . Two hard limitations: $n = 10$ single visits with no repeat sampling gives no within-site variance, and February in Chiang Mai means the regional haze background is confounded with the cooking source — no outdoor or upwind reference was reported. A paired indoor/outdoor low-cost node would have resolved that for a fraction of the cost.

doi: 10.3390/toxics14070643

Park et al. 2026 Sci Total Environ

Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon

Mock-up chamber characterisation of softwood finishes plus field monitoring in an occupied childcare facility and an unoccupied university classroom — before remodelling, 24 h after, and at four weeks — covering formaldehyde, anthropogenic and biogenic VOCs, particulate matter and total airborne bacteria, with a life-cycle global warming potential assessment. α -Pinene dominated the biogenic emission profile. In the occupied, ventilated building PM and bacterial counts fell or stabilised and most VOCs declined within four weeks; in the unoccupied, poorly ventilated classroom **α -pinene rose from 28.7 to 1713.6 $\mu\text{g m}^{-3}$ over four weeks** — a 60-fold accumulation. Net GWP in that building fell to 37.7 kgCO₂eq, 9–11 times below non-wood alternatives, on biogenic carbon storage. With $n = 2$ buildings differing simultaneously in occupancy, ventilation and use, those factors cannot be separated, and the PM measurements are reported only qualitatively. The transferable point is the ventilation dependence: a material acceptable in one air-exchange regime is not in another, which argues for continuous indoor monitoring rather than commissioning-time spot checks.

doi: 10.1016/j.scitotenv.2026.182065

Qi et al. 2026 Ann Work Expo Health

Emission rates of respirable crystalline silica and dust from engineered stone fabrication

Extracts emission rates (mass per time) from four published experimental simulation studies of cutting and grinding engineered stone, tests control technology and task as determinants by regression, then drives a two-zone mathematical model and validates predictions against task-duration field measurements. RCS emission rate rose with slab crystalline-silica content and fell with control technology; **near-field predictions tended to underestimate measured exposures** though the distributions overlapped. Four source studies is a thin evidence base and the underestimation direction is exactly the wrong way for a protective standard. Methodologically the most transferable record in this cluster — an explicit emission-rate-to-exposure model rather than another exposure survey.

doi: 10.1093/annweh/wxag060

Ramsden 2026 J Xenobiot

Xenobiotic hazards in aircraft cabin air

Review of bleed-air contamination in the ~95% of airliners drawing cabin air from turbine compressors: engine-oil traces, tricresyl phosphate, and ultrafine particles abraded from turbine blades. Argues no safe lower TCP limit given inter-individual sensitivity, and presents a life-quality-index affordability calculation. The only ultrafine-focused record in today's corpus.

doi: 10.3390/jox16040119

Subway PSD study 2026 J Environ Sci (China)

PM and CO₂ in Chinese subway stations with varying platform screen door forms

Seven stations instrumented across platform, hall and outdoor locations, contrasting full-height platform screen doors (PSD) with half-height platform bailout doors (HPBD). Platform > hall > outdoor for PM; CO₂ tracks passenger count linearly. The piston effect on train arrival raises **PM_{2.5} by 20% and PM₁₀ by 36% under HPBD**, appreciably more than under PSD, while CO₂ falls 1.8% (PSD) and 4.6% (HPBD). Fine particles carry 38.22–62.57% of total mass, highest behind PSD, and Fe dominates the elemental profile across six apportioned sources. The design is a snapshot without seasonal or diurnal repetition, and the door form is confounded with station age, depth and ventilation design, so the PSD-versus-HPBD contrast is associational. Still the cleanest demonstration in today's corpus that a metric-and-geometry-specific enclosed

microenvironment needs its own sensor siting logic — a platform-edge node and a concourse node are measuring different populations of particles.

doi: 10.1016/j.jes.2025.12.027

Zhou L et al. 2026 Atmos Chem Phys

Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential

Chamber characterisation of Bakhoor under charcoal-assisted heating representative of household practice, with acellular DTT activity and a macrophage intracellular oxidative-stress assay, benchmarked against protocol-matched sidestream cigarette smoke. Normalised per gram burned: **670–1690 $\mu\text{g min}^{-1} \text{g}^{-1}$ mass and $(6-7) \times 10^{11} \text{ particles min}^{-1} \text{g}^{-1}$ number, with ultrafines 70–75% of number and UFP emission rates above sidestream cigarette smoke.** Emission magnitude followed a nonlinear power law in the hexane-soluble fraction, identifying a compositional control. **Water-soluble OP_{DTT} was $\approx 32 \text{ pmol min}^{-1} \mu\text{g}^{-1}$ — modestly below cigarette particles — yet the cellular response was stronger,** and ozone ageing raised OP for both sources at ageing equivalent to days of indoor residence. The acellular/cellular divergence is the finding with teeth: DTT alone would have ranked this source as the safer of the two. Chamber conditions and a charcoal co-source mean the absolute rates are not a household exposure estimate.

doi: 10.5194/acp-26-10587-2026

van Rheeën et al. 2026 J Hosp Infect

Microbial contamination of air-conditioning and humidifying systems in hospitals and the built environment

Mapping review of pathogen presence, biofilm formation and dissemination in the air and water compartments of air-handling systems analogous to adiabatic humidifiers, searched across three databases: **22 of 468 studies retained, covering 184 air-handling units, none of which used adiabatic humidification.** *Pseudomonas aeruginosa* and *Burkholderia cepacia* appeared occasionally in system air; *Mycobacterium* and *Acinetobacter* on cooler surfaces and in drain pans; *Mucor* and *Aspergillus fumigatus* in water and surrounding air, *Fusarium* more airborne. **No included study examined biofilm** — the proposed mechanism is therefore unevidenced within the review's own base. The conclusion is a statement about absence of evidence for the technology actually being adopted on energy-efficiency grounds, which is a legitimate thing to publish. Bioaerosol rather than mass concentration, so it sits at the edge of this watch's scope; included because humidification is a growing indoor-air intervention with no monitoring standard.

doi: 10.1016/j.jhin.2026.07.020

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Yu et al. 2026 Atmos Pollut Res

Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.apr.2026.103162

Burden, policy & mitigation

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 13 RECORDS IN THIS SUBTOPIC)

Abi Nader et al. 2026 Disaster Med Public Health Prep

Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict

Satellite-derived PM_{2.5} over Beirut's southern suburbs compared pre-war (1 Aug–22 Sep 2024) against war (23 Sep–29 Nov 2024), plus ground PM_{2.5} at five sites over ten days. Satellite median rose **15.11 → 20.21 $\mu\text{g}/\text{m}^3$** ; ground median in the direct war zone **19.33 → 22.16 $\mu\text{g}/\text{m}^3$** . Applying BAPHE exposure–response functions gives **RR 1.011 (AR 1.13%) respiratory and RR 1.032 (AR 3.10%) skin** admissions, corroborated by WHO AirQ+. The estimates are transferred, not observed — no admissions data were analysed — so these are projections conditional on a borrowed concentration–response function.

doi: 10.1017/dmp.2026.10412

Hu et al. 2026 Front Public Health

Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants

Two-stage design on the 2015 China 1% Population Sample Survey: Bayesian random-effects smoothing of age-specific mortality for 60+ across 31 provinces, abridged life tables, then a bake-off of MLR, spatial error, spatial lag, GWR and MGWR for socio-environmental determinants. **PM₁₀ was not significant in any global model**; only GWR suggested a more negative association in south-western China, and GWR outperformed MGWR (R^2 0.789 vs 0.664), which the authors correctly attribute to bandwidth instability at $n = 31$. Read this as a negative result about method, not about PM: 31 aggregated provincial units, one cross-section, provincial mean PM₁₀ as the exposure, and no lag structure cannot recover a pollution–longevity gradient that individual-level cohorts detect routinely. The paper is a clean illustration of how ecological aggregation destroys an exposure contrast that a denser monitoring grid would preserve.

doi: 10.3389/fpubh.2026.1851714

Karamzad et al. 2026 Health Sci Rep

Ischaemic heart disease burden in MENA, 1990–2021 (GBD 2021)

21 countries. 2021 age-standardised IHD prevalence 6,404.8/100,000, mortality 202.8, DALY rate 4,023.2. **Ambient PM pollution accounts for 28.2% of IHD DALYs**, third after systolic blood pressure (49.6%) and LDL cholesterol (39.6%). Age-standardised rates declined 1990–2021 but burden remains high and inversely non-linearly related to SDI.

doi: 10.1002/hsr2.72886

Kraus et al. 2026 Int J Hyg Environ Health

A risk-based air quality index for Germany

Rebuilds the German AQI on risk equivalence rather than convention: pollutant–outcome pairs with causal or likely-causal evidence were identified, effect estimates extracted from systematic reviews, daily-mean effects transformed to hourly using German monitoring data, and equivalence coefficients derived by Monte Carlo simulation standardised to PM_{2.5} as reference. PM_{2.5} thresholds anchor to WHO 2021 and the EU information threshold; PM₁₀, NO₂, O₃ and SO₂ thresholds follow by risk equivalence, with an added multi-pollutant module. Evaluated on 2019 and 2022 hourly station data: a marked shift from “very good” to “good”/“moderate” and modestly more “poor” days. The daily-to-hourly effect transformation is the load-bearing and least verifiable assumption, and the index inherits every bias of the source meta-analyses. The most directly policy-relevant record today, and a clean template for making an index defensible rather than traditional.

doi: 10.1016/j.ijheh.2026.114867

Li et al. 2026b Ann Acad Med Singap

Burden trends of Alzheimer disease and other dementias in China, 1990–2023

GBD 2023 secondary analysis, GATHER-compliant. Deaths rose 122,411 → 586,002; DALYs and prevalence more than tripled; age-standardised rates rose only modestly, so the burden growth is demographic. Women carry higher ASRs in 2023 but male rates are rising faster; burden concentrates at ≥75 years. Among the modelled risk factors, **ambient particulate matter accounts for the largest attributable share in 2023**, while household solid-fuel air pollution declined. That attributable fraction is a GBD model output, not an epidemiological estimate from Chinese cohorts, and it inherits the comparative risk assessment's exposure–response assumptions wholesale.

doi: 10.47102/annals-acadmedsg.2026146

Makkar et al. 2026 Environ Sci Pollut Res

Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems

Proposes mycofiltration as a complement to conventional filtration on the argument that HEPA-class media remove particulate mass efficiently but interact weakly with the chemically reactive fraction (aldehydes, PAHs, VOCs, transition metals). Mechanisms offered: extracellular oxidoreductases for aromatic VOC/PAH oxidation, chitin–glucan–melanin wall polymers for metal and soot immobilisation, porous mycelium for interception. Laboratory and pilot-scale evidence only — no field deployment and no bioaerosol-emission safety data. Speculative.

doi: 10.1007/s11356-026-38029-8

Qian et al. 2026 J Hazard Mater

Stronger actions are needed to advance PM_{2.5} mitigation and its health benefits under population aging in China

A new daily 1 km PM_{2.5} product for mainland China 2015–2024, used to track concentration, WHO AQG exceedance and population-weighted exposure, with attributable cardiovascular mortality and a decomposition isolating the ageing offset. National PM_{2.5} fell **35.57 → 20.62 µg/m³** (population-weighted –37.35%), and the area above the WHO interim target of 35 µg/m³ collapsed from 51.07% to 1.49% — **yet over 99% of the population still lives above the 5 µg/m³ guideline**. Attributable CVD deaths fell from 2.16 (1.64–2.58) million in 2015 to 1.55 (1.14–1.91) million in 2022, then **rebounded to 1.76 (1.30–2.15) million in 2024**. The share of health benefit cancelled by population ageing rose from around 40% before 2020 to **56.54% in 2024**, exceeding 70% in some subregions. The burden estimates inherit exposure–response functions from external cohorts, so the uncertainty intervals understate structural uncertainty, and the 1 km product's own error is not propagated into the mortality intervals. The demographic-offset framing is the contribution and it generalises beyond China.

doi: 10.1016/j.jhazmat.2026.143101

Rees et al. 2026 PLoS One

AFRI-c HEPA filtration in care homes: mixed-methods process evaluation

Process evaluation of a cluster RCT across 22 care homes (25 staff, 20 resident and 12 relative interviews; 1,158 resident questionnaires), using normalisation process theory. Self-reported fidelity was high with low contamination — **the trial's null infection result is therefore unlikely to be an adherence artefact**. HEPA units did not change infection-control practice or environmental satisfaction; some residents disliked the draught. An important negative-result anchor for indoor filtration claims.

doi: 10.1371/journal.pone.0347989

Shan et al. 2026 J Environ Sci (China)

Policy-driven emission reductions and post-pandemic rebound in China

High-resolution integrative inventories for 2018, 2020 and 2023 covering CO₂ and nine pollutants. In 2020 vs. 2018, SO₂ (–18.1%), VOCs (–32.6%) and CO (–8.8%) fell while CO₂ (+11.6%), NO_x (+16.8%) and PM (+21.9%) rose on expanded power generation and metal production. 2023 saw 3.2–15.1% rebound across pollutants. The Fenwei Plain shows persistent PM growth. Useful for anyone building policy-counterfactual exposure scenarios.

doi: 10.1016/j.jes.2026.01.009

Sun et al. 2026 Br J Cancer

Clarifying causal and translational inference in studies of ambient PM_{2.5} and lung cancer

Correspondence, not primary research, arguing that the PM_{2.5}–lung-cancer literature conflates aetiological causal claims with translational or prognostic ones, and that the distinction changes what an estimate licenses. No new data. Worth keeping in the register because the same conflation runs through exposure-assignment practice: a satellite- or LUR-derived annual mean supports population-attributable-fraction arithmetic but not the individual-level counterfactual that translational framing implies, and adding a denser sensor network narrows the measurement error without touching the identification problem.

doi: 10.1038/s41416-026-03555-2

Vellaiyan 2026 J Environ Health Sci Eng

Diesel NO_x emissions and public health: a source–pathway–receptor synthesis

Review spanning combustion science, atmospheric chemistry, exposure assessment and epidemiology — disciplines that normally publish past each other. Useful on near-road gradients, street-canyon trapping, and the gap between certification cycles and real-world driving emissions. A synthesis, not new evidence; the epidemiological half is thinner than the combustion half.

doi: 10.1007/s40201-026-00995-z

Zhou et al. 2026b Toxics

Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis

Spatial Durbin model over the 28 “2+26” Beijing–Tianjin–Hebei channel cities, relating pollution-intensive industrial activity to haze, and attributing ultrafine particle generation to desulfurisation, denitrification and dust-removal operations — specifically lax wet-FGD management, ratcheting NO_x standards, and electrostatic precipitators that are ineffective in the ultrafine size range. Direct effect 0.028, total spillover-inclusive effect 0.151. **No ultrafine particles were measured anywhere in this study**. The UFP attribution is inferred from city-level industrial statistics and a haze proxy, so the central claim — that abatement equipment is itself a UFP source — is an untested hypothesis dressed as a finding, however physically plausible the ESP argument is. Logged tier C for the hypothesis, not the evidence; it is exactly the claim a size-resolved mobility-sizer campaign near a controlled stack could settle.

doi: 10.3390/toxics14070588

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Faggi et al. 2026 Atmos Environ

Heatwaves and particulate matter increases: an Italian case study

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122246

Other clinical endpoints

10 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 10 RECORDS IN THIS SUBTOPIC)

Doğan Karataş et al. 2026 J Clin Med*Air pollution and sudden sensorineural hearing loss: nine-year retrospective study*

223 SSHL patients (mean 52.7 ± 7.4 y) at a single ENT clinic in Sivas, Türkiye, 2016–2025, with monthly mean PM_{10} , $PM_{2.5}$, SO_2 , NO_x , NO , CO and NO_2 from ministry monitoring stations and a “cumulative risk” computed by a public-health specialist. Post-treatment audiometric change was significant at all frequencies (greatest at 2 kHz) and no significant seasonal variation in case occurrence was found. The design cannot answer the question it poses: a case series with no control period and no non-case comparison cannot estimate an exposure–occurrence association, and pairing 223 events with monthly city-wide pollutant means guarantees exposure misclassification far exceeding any plausible effect. The audiometric outcomes are a treatment-response analysis with no pollution content. Included for completeness of the register.

doi: 10.3390/jcm15145687

He et al. 2026 Toxics *$PM_{2.5}$ -bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator*

Largest study in today’s corpus: 122,965 valid questionnaires from school-aged children across the Pearl River Delta, with $PM_{2.5}$ -bound chlorinated paraffin exposure analysed by generalised linear mixed models, restricted cubic splines, WQS regression and quantile g-computation, plus mediation analysis. Per-IQR Σ CP exposure raised sleep-disorder odds with **ORs reported as a 1.08–3.20 range**; short-chain CPs were the dominant contributor; body weight mediated 3.40–35.67%; associations were stronger in girls and in children under 12. Two serious problems. The exposure is almost certainly modelled or area-assigned rather than personally measured — CP speciation on $PM_{2.5}$ is a specialist analysis not deployable at $n > 10^5$ — so the exposure contrast is spatial and confounded with everything else that varies across the PRD. And an OR range spanning 1.08–3.20 without a stated model set is uninterpretable: it could be lag structure, subgroup, or CP congener. Large n does not fix either.

doi: 10.3390/toxics14070607

Ion-Andrei et al. 2026 Biomedicines*Chronic air pollution exposure and the immunomolecular profile of lung cancer*

Regional retrospective series in Constanța County, Romania, linking ≈ 15 -year historical monitor-based pollution exposure to tumour histology, PD-L1 expression and EGFR status, with ALK/KRAS described only. **Chronic NO_2 was associated with higher PD-L1 expression (OR 1.035, $p = 0.017$)**, with PM_{10} , NO and NO_x trending in the same direction and an exploratory CO –EGFR signal. The idea — that ambient exposure shapes tumour immunophenotype and therefore immunotherapy eligibility — is genuinely interesting and underexplored. The execution does not support it: no CI or sample size in the abstract, no adjustment for smoking described, an exposure assigned from county monitors over 15 years with no residential-history weighting, and multiple correlated pollutants tested without correction. Hypothesis-generating at best, and the authors concede causality cannot be established.

doi: 10.3390/biomedicines14071615

Knapova et al. 2026 Appl Psychol Health Well Being*Daily relationship between air pollution and physical activity in an industrial region*

693 adults in the industrial Ostrava region of the Czech Republic, 9,702 person-days, device-measured step counts against daily PM_{10} over 14 days, multilevel Bayesian models with within-person centring and adjustment for temperature, sunshine and precipitation. On days when PM_{10} exceeded a participant’s own 14-day mean, step counts fell ($b = -0.027$, 95% CI -0.047 to -0.006); **a 1-SD PM_{10} increase corresponded to 361 fewer steps**. Self-reported tendency to monitor air quality did not moderate the effect, which argues the response is not deliberate avoidance behaviour. Exposure is ambient monitor data assigned to residence, not personal, so the estimate confounds pollution with unmeasured weather and visibility cues; 14 days is short for a within-person design; and non-probabilistic quota sampling limits transport. The finding matters for exposure modelling regardless: **behavioural response makes time-activity patterns endogenous to concentration**, which biases any exposure model that treats mobility as exogenous.

doi: 10.1111/aphw.70186

Lupita et al. 2026 Dent J *$PM_{2.5}$, preventive-care capacity and oral disease burden across the EU*

Ecological panel across 27 EU member states, 2020–2023, GBD outcomes vs. national mean $PM_{2.5}$ with unmet dental care need as an effect modifier. $PM_{2.5}$ was *not* independently associated with caries after adjustment, but the $PM_{2.5} \times$ unmet-need interaction was significant; $PM_{2.5}$ was independently associated with edentulism. Ecological fallacy risk is high and the authors say so.

doi: 10.3390/dj14070440

Ouyang et al. 2026 J Hazard Mater*Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities*

879,356 daily eczema outpatient records from two Guangdong dermatology hospitals matched to daily $PM_{2.5}$, PM_{10} , NO_2 , SO_2 and O_3 ; distributed lag non-linear models for single pollutants and quantile g-computation for two four-pollutant mixtures. Per $10 \mu g/m^3$, **SO_2 gave the largest single-pollutant effect at +22.63%**. Each quartile increase in the PM_{10} -based mixture raised visits 4.14% (49.5% of the weight on PM_{10}); the $PM_{2.5}$ -based mixture raised them 4.21%, but with the weight on SO_2 (38.1%) and NO_2 (37.5%). Both mixtures correspond to ≈ 18 extra visits per day. Outpatient visits confound severity with care-seeking behaviour, and no confidence intervals are reported for the mixture estimates.

doi: 10.1016/j.jhazmat.2026.143124

Park et al. 2026 JAAD Int*Long-term particulate matter exposure and risk of androgenetic alopecia*

Nationwide Korean cohort research letter linking long-term $PM_{2.5}/PM_{10}$ to androgenetic alopecia incidence, with hair-follicle inflammation and oxidative stress as the proposed mechanism. Abstract not indexed; full text required for effect sizes.

doi: 10.1016/j.jdin.2026.06.008

Sun et al. 2026 J Neurosurg Pediatr*Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes*

881 patients, 2008–2023 single-institution registry, geocoded to Area Deprivation Index and CDC/ATSDR environmental measures plus satellite-derived ground-level $PM_{2.5}$. Multivariable Cox with cluster-robust SEs: high ozone HR 1.38 (1.05–1.82) and high ADI HR 1.46 (1.15–1.86) for shunt failure; high coal exposure HR 1.92 (1.28–2.88) for shunt infection; **high $PM_{2.5}$ HR 3.16 (1.11–8.99) for all-cause mortality**. The mortality CI is very wide — treat as signal, not estimate.

doi: 10.3171/2026.3.PEDS2615

Zhou et al. 2026 BMJ Open*Environmental exposome study for prostate cancer in elderly men from Western China*

580 histologically confirmed localised prostate cancers after radical prostatectomy, matched 1:2 to 1,160 cancer-free controls from the WeEndPd cohort, with urinary bisphenols and phthalate metabolites by LC-MS/MS and geocoded five-year SO₂ and PM_{2.5}. Highest-quartile five-year SO₂ (9.37–28.79 µg/m³) gave **OR 1.65 (95% CI 1.08–2.72)** against the lowest; MECPP, MEHHP, MEHP, BPA and BPZ were all higher in cases; the WQS mixture of endocrine-disrupting chemicals gave **OR 1.26 (1.20–1.33)**. The PM_{2.5} term is reported but is not the headline. Two structural problems: urinary EDC metabolites have half-lives of hours to days and were measured at or after diagnosis, so they cannot index the aetiologically relevant window — this is reverse-causation-prone by construction; and cases are surgical patients from one tertiary centre, which selects on operability and on health-seeking behaviour that also tracks residential air quality.

doi: [10.1136/bmjopen-2025-110957](https://doi.org/10.1136/bmjopen-2025-110957)

Zurita et al. 2026 Pediatr Blood Cancer*Childhood leukaemia and proximity to benzene emission sources in the Mexico City area*

Ecological study of acute lymphoblastic leukaemia cases from one paediatric hospital, 2010–2021, against counts of benzene-emitting establishments (petroleum and coal product manufacture, fuel and lubricant retail, printing, chemical industry) within distance buffers, using multivariate spatial models by municipality, sex and age group. Significant positive association with source counts for ages 2–17. Single-hospital case ascertainment plus municipality-level exposure is a textbook ecological design: no individual exposure, no residential history, and referral patterns to one hospital are almost certainly spatially structured. Hypothesis-generating at best — the authors say as much.

doi: [10.1002/1545-5017.70482](https://doi.org/10.1002/1545-5017.70482)

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Cai et al.	PM _{2.5} as the dominant air pollutant associated with suicide among people living with HIV/AIDS: province-wide case-crossover	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Calderon-Garcidueñas	Abnormal eye movements reflect early cortical and volumetric brain changes in PM _{2.5} -exposed urban youth	<i>J Alzheimers Dis</i>	Cross-sectional imaging	PM _{2.5}	Mexico
Cattarinussi	Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories	<i>medRxiv</i>	Prospective cohort	PM _{2.5}	USA
Dales	Wildfire-sourced PM _{2.5} and ED visits for migraine and other primary headache	<i>JAMA Netw Open</i>	Case-crossover	PM _{2.5} (wildfire)	Canada
Daoud et al.	Air pollution and suicide mortality in the UK: implications for environmental and mental health policy	<i>Environ Sci Pollut Res Int</i>	Ecological panel, fixed effects	PM _{2.5} (modelled, DEFRA)	UK
Glover	Umbrella meta-review of urban environmental exposures and cognitive health	<i>Age Ageing</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Goossens	In utero air pollution exposure and pre-/postnatal brain development	<i>Neurosci Appl</i>	Narrative review	PM _{2.5} / PM ₁₀ / PM _{coarse}	Global
Kabir	Neighbourhood socioeconomic inequalities in mental health: moderation by environmental attributes	<i>Health Place</i>	Scoping review	PM (unspecified) / emissions	Global
Lu	Subchronic PM _{2.5} exposure exacerbates chronic stress-induced depression-like pathology in mice	<i>Environ Res</i>	Murine inhalation exposure	PM _{2.5}	China
Ma	PM _{2.5} -induced neurotoxicity and depression: exposure risks and mechanisms	<i>iScience</i>	Narrative review	PM _{2.5} + components	Global
Pan	Maternal PM exposure and offspring neurodevelopment: effects and mechanisms	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Narrative review	PM (bulk)	China
Sarkar	Particulate-matter-induced neurotoxicity: neuroinflammation and organelle dysfunction	<i>J Appl Toxicol</i>	Narrative review	PM _{2.5} + PM ₁₀ jointly	Global
Zhang et al.	Association of PM _{2.5} and its components with cognitive dysfunction among rural older adults	<i>Neurotoxicology</i>	Cross-sectional (cohort baseline)	PM _{2.5} , BC, OM, NH ₄ , SO ₄ , NO ₃ (TAP)	China
Cardiovascular & metabolic					
Belay	Long-term bushfire-related PM _{2.5} and coronary plaque burden in asymptomatic adults	<i>Eur J Prev Cardiol</i>	Prospective cohort	PM _{2.5} (bushfire)	Australia
Chen	Long-term particulate matter exposure and coronary inflammation by pericoronary fat attenuation index	<i>Environ Pollut</i>	Retrospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Chrysohoou	Preventing cardiovascular disease within the built and living environment: an expert opinion	<i>Eur J Prev Cardiol</i>	Narrative review	PM (unspecified)	EU-27
Huang	PM _{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Cross-sectional (multistage)	PM _{2.5}	China
Kazmierska	Air pollution and cardiovascular disease: review of current data	<i>Przegl Epidemiol</i>	Narrative review	PM _{2.5} / PM ₁₀	Global
Luo	Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning	<i>Front Public Health</i>	Time-series	PM _{2.5} + PM ₁₀	China
Rusconi	Long-term ultrafine particle exposure 2010-2019 in high-income countries and cardiovascular burden	<i>GeoHealth</i>	Ecological panel	Ultrafine particles	Global
Xi	Air pollution, built environment and temperature vs. prediabetes phenotypes	<i>Diabetologia</i>	Prospective cohort	PM _{2.5}	Germany
Yang	Short-term PM _{2.5} and hospitalisation/ED use in atrial fibrillation	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} (wildfire-influenced)	USA
Zhao	Compound exposure to extreme heat and substantial landscape fire-sourced air pollution and cardiovascular mortality	<i>J Am Coll Cardiol</i>	Case-crossover	PM _{2.5} (fire-sourced)	Global
Zhou	Air pollution and hypertension: umbrella review of systematic reviews and meta-analyses	<i>iScience</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀	Global
Reproductive & developmental					
Bello et al.	Prenatal exposure to PM _{2.5} and infant health: evidence from Quebec	<i>Econ Hum Biol</i>	Registry cohort, FE models	PM _{2.5} (monitored assigned)	Canada

First author	Focus	Journal	Design	Metric	Region
Chen	Prenatal air pollution, fetal growth trajectory phenotypes and child neurodevelopment	<i>J Environ Sci (China)</i>	Prospective cohort	PM _{2.5} / PM ₁₀	China
Huang	Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures)	PM _{2.5} / PM ₁₀	China
Li	Ambient air pollutant exposure and semen quality: a refined multi-window panel study	<i>Ecotoxical Environ Saf</i>	Panel study	PM _{2.5} + PM ₁₀	China
Lin	Maternal particulate matter exposure during pregnancy and offspring liver function	<i>Environ Pollut</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Liu	Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models	<i>Int J Mol Sci</i>	In vitro (cell lines)	PM _{2.5}	Netherlands
Pai et al.	Household incense burning and small for gestational age risk in newborn infants	<i>Pediatr Neonatol</i>	Prospective birth cohort	Incense PM (indoor, unspecified)	Taiwan
Peterson	Ambient air pollution, PAHs and pubertal development: critical review	<i>Curr Environ Health Rep</i>	Critical review	PM _{2.5} components	Global
Sezer et al.	Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate	<i>Birth Defects Res</i>	Case-control	PM _{2.5} + PM ₁₀ + O ₃	Turkiye
Xie	Long-term PM _{2.5} and timing of pubertal development in boys and girls	<i>Environ Res</i>	Prospective cohort	PM _{2.5}	China
Respiratory & allergic					
Bayar Muluk	Climate change and otolaryngology: airway disease impacts and emission reduction	<i>Int J Biometeorol</i>	Narrative review	PM _{2.5}	Global
Chaiwong	PM _{2.5} and systemic inflammatory biomarkers in elderly COPD patients versus healthy elderly	<i>Int J Mol Sci</i>	Panel study	PM _{2.5}	Thailand
Ghana AIRCLARE	Short-term ambient PM _{2.5} /PM ₁₀ and acute respiratory morbidity in Ghana: time-stratified case-crossover pilot	<i>Research Square (preprint)</i>	Case-crossover	PM _{2.5} / PM ₁₀	Ghana
Li	Human genetic variation, airway microbiome, environmental exposure and respiratory health	<i>mSystems</i>	Prospective cohort	PM _{2.5} only	China
Liao	Short-term air pollution, respiratory symptoms and lung function in asthma	<i>Med Gas Res</i>	Panel study	PM _{2.5} / PM ₁₀	China
Lu et al.	Antagonistic effects of PM _{2.5} and ozone on respiratory morbidity: multi-pollutant interaction, Pearl River Delta	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Mokoena	Socio-economic conditions, air pollution sources and eczema symptoms among teenagers in Soshanguve	<i>Public Health Chall</i>	Cross-sectional (multistage)	PM (unspecified) / emissions	South Africa
Sanchez de Prada	Ambient PM and asthma-coded admissions in children 0-3 y, Spain	<i>Epidemiologia</i>	Ecological panel	PM _{2.5} / PM ₁₀	Spain
Wang	Environmental and socioeconomic influences on childhood asthma and pneumonia burden	<i>Front Med</i>	Retrospective cohort	PM _{2.5}	China
Yang	Allergenic vs. toxic risk of airborne seed hairs compared with ambient PM _{2.5}	<i>J Hazard Mater</i>	Field measurement	PM _{2.5} + bioaerosol	China
Yue et al.	Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou	<i>Front Public Health</i>	Time-series	Traffic gases (NO ₂ , SO ₂)	China
Zhang	Air pollution exposure in infancy and incidence of atopic dermatitis in preschool-aged children	<i>Epidemiology</i>	Prospective cohort	PM _{2.5} only	USA
Zhu et al.	Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis	<i>Medicine (Baltimore)</i>	Time-series	PM _{2.5} + PM ₁₀ + 4 gases	China
Mechanistic toxicology					
Balu et al.	SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction	<i>J Biochem Mol Toxicol</i>	Ex vivo perfused organ	Diesel PM (SRM 1650b)	India
Ding	PM _{2.5} promotes vascular calcification via miR-27b-5p/HES1 signalling	<i>J Environ Sci (China)</i>	In vitro + organoid + murine	PM _{2.5}	China
Fakfum	Seasonal PM _{2.5} exposure and plasma metabolome changes related to metabolic syndrome	<i>Toxics</i>	Repeated-measures biomarker	PM _{2.5}	Thailand
Liu	Resveratrol ameliorates PM _{2.5} -aggravated IBS-D via gut bacteria and hypoxanthine/linoleic acid metabolism	<i>Environ Toxicol Pharmacol</i>	Murine inhalation exposure	PM _{2.5} only	China
Lu	Toxicity of PM _{2.5} from incomplete solid-fuel burning in <i>Caenorhabditis elegans</i>	<i>Toxics</i>	Controlled exposure (ecotox)	PM _{2.5} (biomass)	China

First author	Focus	Journal	Design	Metric	Region
Maraschi	Osmoregulatory disruption and metal bioaccumulation from settleable atmospheric PM	<i>J Comp Physiol B</i>	Controlled exposure (ecotox)	Settleable APM	Brazil
Mishra et al.	Aerosol oxidative potential and reactive species predicted with a chemical kinetics model	<i>Atmos Chem Phys</i>	Kinetic model + field validation	PM composition, OP-DTT / OP-AA	France / UK
Pan et al.	Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis	<i>Front Pharmacol</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀ / DEP (animal)	Global
Saygin et al.	Source identification and seasonal change of trace metals, protein content and oxidative potential of fine and coarse particulates after the Kahraman-maras earthquakes	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Turkiye
Wang	PM exposure, mitochondrial stress and extracellular matrix remodelling in scleral fibroblasts	<i>Exp Eye Res</i>	In vitro + organoid + murine	PM (unspecified) / emissions	China
Wu	Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation	<i>J Environ Sci (China)</i>	Real-ambient murine + co-culture	Real-ambient PM	China
Occupational & indoor					
Chen	Nested suction instrument to cut intraoperative surgical-smoke PM _{2.5}	<i>Front Med Technol</i>	Bench / device experiment	PM _{2.5} (surgical smoke)	China
Cole	Determinants of respirable crystalline silica exposure in Australian tunnelling	<i>Ann Work Expo Health</i>	Observational exposure study	Respirable dust / RCS	Australia
Ding	Stage-resolved hair Zn-isotope dynamics as a firefighting combustion-exposure tracer	<i>J Hazard Mater</i>	Repeated-measures biomarker	Combustion-derived PM	China
Ghosh	Respiratory quality of life and pulmonary function among traffic police in eastern India	<i>Indian J Occup Environ Med</i>	Cross-sectional (multistage)	PM (unspecified) / emissions	India
Gwak	Size-resolved modelling of airborne viral RNA copies in indoor environments	<i>Risk Anal</i>	Physical exposure model	Bioaerosol / coarse	Global
Komonnithiphong	PM _{2.5} -bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants	<i>Toxics</i>	Field measurement	PM _{2.5} components	Thailand
Park et al.	Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon	<i>Sci Total Environ</i>	Field measurement	PM (indoor, unspecified)	South Korea
Qi	Emission rates of respirable crystalline silica and dust from engineered stone fabrication	<i>Ann Work Expo Health</i>	Bench / device experiment	Settleable / respirable dust	Global
Ramsden	Xenobiotic hazards in aircraft cabin air	<i>J Xenobiot</i>	Review + risk economics	Ultrafine particles	Global
Subway PSD	Characterisation and source apportionment of PM and CO ₂ in subway stations with differing platform door forms	<i>J Environ Sci (China)</i>	Field measurement	PM _{2.5} / PM ₁₀	China
Yu et al.	Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Zhou L. et al.	Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential	<i>Atmos Chem Phys</i>	Chamber emission characterisation	UFP number + PM mass, DTT/cellular OP	Chamber (Gulf practice)
van Rheenen et al.	Microbial contamination of air conditioning and humidifying systems in hospitals and the built environment	<i>J Hosp Infect</i>	Mapping review	Bioaerosol (bacteria, fungi, protozoa)	Multi-country
Sensing, forecasting & instrumentation					
Aher	Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy	<i>Environ Sci Pollut Res</i>	Commentary / methods letter	PM (unspecified)	India
Ainembabazi	Trends in atmospheric pollution research in Uganda, 1990-2025: a scoping review	<i>Toxics</i>	Scoping review	PM _{2.5} / PM ₁₀	Uganda
Feng	Calibration strategies and measurement frameworks for low-cost particulate matter sensors: a review	<i>Meas Sci Technol</i>	Critical review	PM _{2.5} only	Global
Ginsburg	Global and regional calibration of low-cost PM _{2.5} sensors (PurpleAir): transferability of US-based corrections	<i>Aerosol Sci Technol</i>	Field measurement	PM _{2.5} only	Global
Huda	Low-cost sensor network for wildfire monitoring and air quality at remote community locations	<i>J Air Waste Manag Assoc</i>	Field measurement	PM _{2.5} only	Canada
Kumpika et al.	Correction of humidity and sensor aging effects in low-cost PM _{2.5} light-scattering sensors	<i>Environ Pollut Bioavailab</i>	Field co-location + chamber	PM _{2.5} (PMS7003 vs BAM)	Thailand
Le et al.	Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling	<i>Atmos Meas Tech</i>	Multi-instrument field evaluation	Attenuated backscatter, depolarisation ratio	Finland / Germany

First author	Focus	Journal	Design	Metric	Region
Mohapatra et al.	IoT-machine learning proof-of-concept framework for real-time urban particulate matter prediction	<i>IEEE Access</i>	Proof-of-concept deployment	PM ₁ / PM _{2.5} / PM ₁₀ (PMS7003)	India
Msigwa	AIoT real-time sensing and calibration of low-cost PM sensors using the TCM network	<i>IEEE Internet Things J</i>	Deep-learning model	PM _{2.5} only	Global
Nurseitov et al.	Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter	<i>Information</i>	Network deployment + field evaluation	PM _{2.5} / PM ₁₀ (ZPHS01B)	Kazakhstan
Park et al.	Diagnosing explainability of machine learning: transformer-based PM _{2.5} estimation in Seoul	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Republic of Korea
Patel	Scalable air-quality sensor placement via gradient-based mutual information maximization	<i>Proc AAAI</i>	Deep-learning model	PM _{2.5} only	Global
Patton	Human health risk of particle exposure in outdoor environments in southwestern Pennsylvania	<i>Environ Sci Pollut Res</i>	Real-time sensor campaign	PM _{2.5} only	USA
Qian et al.	Enhancing accuracy of indoor air quality sensors via automated machine learning calibration	<i>Atmos Meas Tech</i>	Chamber co-location, AutoML calibration	PM _{2.5} (two LCS models vs reference)	Chamber
Reis	Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference	<i>Sensors</i>	Simulation (system-level)	PM + gas, multi-parameter node	Simulation
Shi	Adaptive spatiotemporal graph transformer for multi-site multi-step PM _{2.5} forecasting	<i>J Environ Manage</i>	Deep-learning model	PM _{2.5}	China / India
Silva et al.	Low-cost sensors for indoor air quality monitoring: systematic review of accuracy, applications and limitations	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	PM (OPC) and CO ₂ (NDIR)	Multi-country
Tsuchiya	Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices	<i>Biosensors</i>	Real-time sensor campaign	Bioaerosol / coarse	Japan
Wang	Joint monitoring and early warning of SARS-CoV-2 in outdoor PM _{2.5} and wastewater	<i>Viruses</i>	Environmental surveillance campaign	PM _{2.5} (bioaerosol)	China
Zheng et al.	Augmented transformer framework for PM _{2.5} chemical composition via one-step LDI single-particle aerosol mass spectrometry	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Exposure assessment & modelling					
Blanco et al.	Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences	<i>J Expo Sci Environ Epidemiol</i>	Design resampling + cohort analysis	UFP particle number concentration	United States
Chen	Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities	<i>J Hazard Mater</i>	Source apportionment	PM _{2.5}	China
Chen et al.	Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability	<i>J Hazard Mater</i>	Exposome-wide association	PM _{2.5} constituents + VOCs	USA
Chengdu NO ₃	Gas-phase reactions accelerate nitrate formation in spring and early summer in a Sichuan Basin megacity	<i>J Environ Sci (China)</i>	Supersite observation	PM _{2.5} components	China
Delgado Peralta et al.	Improving PM _{2.5} simulations in metropolitan Sao Paulo via PMF-based emission inventory adjustment	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Brazil
Ding	Urban-rural balanced estimation framework for PM _{2.5} exposure assessment across China	<i>Geo-spat Inf Sci</i>	Deep-learning model	PM _{2.5} only	China
Hu	Fusion of ground monitoring, satellite observation and chemical transport modelling for seamless hourly 1-km PM _{2.5}	<i>Geo-spat Inf Sci</i>	Deep-learning model	PM _{2.5} only	China
Nguyen	Satellite-driven prediction of PM _{2.5} : machine learning and explainable AI	<i>Eng Res Express</i>	Critical review	PM _{2.5} only	Global
Nyoni et al.	Performance of road-traffic-based exposure proxies against personal PM _{2.5} measurements in three Sub-Saharan African countries	<i>Air Qual Atmos Health</i>	Proxy validation vs personal exposure	PM _{2.5} (personal, filtered)	The Gambia / Kenya / Mozambique
Pajak	Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk	<i>Toxics</i>	Dispersion model + soil sampling	PM ₁₀ / PAH	Poland
Patwary	Spatio-temporal assessment of PM ₁₀ and meteorological correlates in Cumilla, Bangladesh	<i>Res Sq (preprint)</i>	Field measurement	PM ₁₀	Bangladesh
Raj	Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh	<i>Res Sq (preprint)</i>	Field measurement	PM _{2.5} + PM ₁₀	India
Saeedi et al.	Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles	<i>Environ Sci Technol</i>	LUR model comparison	Ultrafine particle number (mobile)	Canada

First author	Focus	Journal	Design	Metric	Region
Sanchez	Morbidity related to level-two eruption of Popocatepetl in communities near the crater	<i>Indian J Occup Environ Med</i>	Ecological panel	PM _{2.5} + PM ₁₀ jointly	Mexico
Solorzano-Araque	Multiscale exposure modelling: LOTOS-EUROS + mobility surveys + particle toxicity	<i>GeoHealth</i>	CTM + mobility integration	PM _{2.5} / PM ₁₀	Colombia
Zhang	High-resolution aerosol liquid water content across the contiguous US using machine learning	<i>npj Clim Atmos Sci</i>	Machine-learning model	PM _{2.5}	USA
Zhou M et al.	Surface PM _{2.5} air pollution in 2022 India: emission updates, WRF-Chem evaluation, and source attribution	<i>Atmos Chem Phys</i>	CTM simulation + source attribution	PM _{2.5} (WRF-Chem, speciated)	India
Burden, policy & mitigation					
Abi Nader	Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict	<i>Disaster Med Public Health Prep</i>	Environmental surveillance campaign	PM _{2.5}	Lebanon
Faggi et al.	Heatwaves and particulate matter increases: an Italian case study	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Italy
Hu	Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants	<i>Front Public Health</i>	Bayesian + geospatial regression	PM ₁₀	China
Karamzad	Ischaemic heart disease burden in MENA 1990-2021; PM attributable fraction 28.2%	<i>Health Sci Rep</i>	GBD secondary analysis	Ambient PM	MENA
Kraus	Development and evaluation of a risk-based air quality index for Germany	<i>Int J Hyg Environ Health</i>	Bayesian + geospatial regression	PM _{2.5} + PM ₁₀ jointly	Germany
Li	Burden trends of Alzheimer disease and other dementias in China, 1990-2023	<i>Ann Acad Med Singap</i>	GBD secondary analysis	PM (unspeciated)	China
Makkar	Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems	<i>Environ Sci Pollut Res</i>	Critical review	PM _{2.5}	India
Qian et al.	Stronger actions are needed to advance PM _{2.5} mitigation and its health benefits under population aging in China	<i>J Hazard Mater</i>	Machine-learning model	PM _{2.5}	China
Rees	HEPA filtration to reduce respiratory infection in care homes	<i>PLoS One</i>	RCT process evaluation	Filtered PM	UK
Shan	Policy-driven emission reductions and post-pandemic rebound, high-resolution inventories	<i>J Environ Sci (China)</i>	Emission inventory	PM emissions	China
Sun	Clarifying causal and translational inference in studies of ambient PM _{2.5} and lung cancer	<i>Br J Cancer</i>	Commentary / methods letter	PM _{2.5}	Global
Vellaiyan	Diesel NOx emissions and public health: source-pathway-receptor synthesis	<i>J Environ Health Sci Eng</i>	Narrative review	PM (unspeciated)	Global
Zhou	Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis	<i>Toxics</i>	Spatial econometric model	Ultrafine particles	China
Other clinical endpoints					
Dogan Karatas	Air pollution and sudden sensorineural hearing loss: nine-year retrospective study	<i>J Clin Med</i>	Retrospective clinical series	PM ₁₀ / PM _{2.5}	Turkiye
He	PM _{2.5} -bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator	<i>Toxics</i>	Cross-sectional (multistage)	PM _{2.5} components	China
Ion-Andrei	Chronic air pollution exposure and the immunomolecular profile of lung cancer	<i>Biomedicines</i>	Retrospective cohort	PM ₁₀ (with NO ₂ /NO _x)	Romania
Knapova et al.	Daily relationship between air pollution and physical activity in an industrial region	<i>Appl Psychol Health Well Being</i>	Panel study	PM ₁₀	Czech Republic
Lupita	PM _{2.5} , preventive-care capacity and caries/edentulism burden across the EU	<i>Dent J</i>	Ecological panel	PM _{2.5}	EU-27
Ouyang	Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities	<i>J Hazard Mater</i>	Time-series	PM _{2.5} + PM ₁₀	China
Park	Long-term PM exposure and risk of androgenetic alopecia	<i>JAAD Int</i>	Nationwide cohort	PM _{2.5} / PM ₁₀	South Korea
Sun	Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes	<i>J Neurosurg Pediatr</i>	Registry cohort + geospatial	PM _{2.5} (satellite-derived)	USA
Zhou et al.	Environmental exposome study for prostate cancer in elderly men from Western China	<i>BMJ Open</i>	Case-control	PM _{2.5} + SO ₂	China
Zurita	Childhood leukaemia and proximity to benzene emission sources in the Mexico City area	<i>Pediatr Blood Cancer</i>	Ecological panel	PM (unspeciated)	Mexico

Provenance and method

This rollup is an **aggregation only**. It pools `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over 27 July 2026–1 August 2026 via `PMRW_START/PMRW_END` and **re-queries no API**, so every record was screened and deduplicated in the daily issue that first carried it. Per-record provenance, source-query windows, exclusions and known metadata defects are documented in those daily issues and are not restated here.

Daily issues pooled: 27, 28, 29, 30, 31 July and 1 August 2026 — six consecutive issues, no gap.

Why the window starts on 27 July and not 26 July. The weekly cadence is Sunday-to-Saturday, so the nominal window is 26 July–1 August. The 26 July slot in `state/corpus/` does not hold a 26 July issue: it holds the **1–26 July retrospective backfill** (164 records, entry window 2026-07-01 -> 2026-07-26) harvested for the July monthly rollup. Pooling it here would put a month of records inside a week, inflate every count by a factor of 2.3 and double-count material already published in `Reports/monthly/PM-Research-Watch-Monthly_2026-07-31.pdf`. It is therefore excluded and the pooled window is stated as 27 July–1 August throughout. The 164 backfill records are not lost — they are in the July monthly in full.

Effect estimates are transcribed from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption; **no meta-analytic summary is computed and no I^2 is reported**, because the estimates are not exchangeable. Trend and inflection statements are judgements over a six-day window and are not statistical tests. The “where this is heading” box is explicitly judgement and is labelled as such. Content derived from PubMed, Europe PMC and Crossref metadata; abstracts remain the copyright of their respective publishers.